Psychometric Evaluation of the 8-Item Patient Health Questionnaire Among Filipino American Emerging Adult Men and Women

Online Supplement: Reproducible Quarto Workflow (Code Appendix)

Delwin B. Carter   Scott Plunkett   David Alpizar   Andrew Ainsworth
Luciana Laganá

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To render to PDF use this code (this will be removed)

quarto::quarto_render(

“PHQ-8 Filipino Americans.qmd”,

output_format = “pdf”)

The complete Quarto workflow and rendered HTML supplement are archived in a Git repository at GitHub (https://github.com/delwincarter/Filipino-PHQ-JPBA) and will be made public upon publication of the manuscript.

Introduction

This online supplement documents a transparent and fully reproducible workflow for the analyses reported in *Psychometric Evaluation of the 8-Item Patient Health Questionnaire Among Filipino American Emerging Adult Men and Women*. The supplement is intentionally code-forward and presents the complete analytic pipeline—including data preparation, scoring, confirmatory factor analyses (CFA), and measurement invariance testing—within a single executable Quarto document.

Purpose and Scope

The supplement is designed to:

1. document all analytic decisions in a structured, step-by-step workflow;
2. enable exact reproduction of model estimation and manuscript tables (given the same input data and software versions); and
3. support methodological review by providing the full lavaan model syntax used to estimate CFA and measurement invariance models across analytic subgroups.

This document is not a narrative duplicate of the manuscript. Instead, it functions as a computational appendix that prioritizes clarity, traceability, and reproducibility.

Although this supplement reproduces the analyses reported in the manuscript, the workflow is written to be reusable. Researchers may adapt the pipeline to other datasets by replacing the input data file and updating subgroup definitions while retaining the CFA, invariance, and reporting structure.

Software and Reproducibility Expectations

All analyses were conducted in **R** using the **lavaan** package to estimate CFA and multigroup measurement invariance models using WLSMV estimation with polychoric correlations for ordinal indicators. The workflow is organized into discrete sections that mirror the analytic sequence reported in the manuscript (data preparation → subgroup definition → CFA estimation → invariance testing), allowing readers to reproduce each stage of the analysis directly from the Quarto file.

All CFA and multigroup invariance models in this supplement are estimated using polychoric correlation matrices under WLSMV estimation in lavaan. Pearson correlations were not used for model estimation

> **PHQ-8 Items and Response Scale**
>
> All analyses in this supplement use the eight indicators from the Patient Health Questionnaire-8 (PHQ-8), which assess the frequency of depressive symptoms over the past two weeks. Respondents are asked:
> "Over the last two weeks, how often have you been bothered by any of the following problems?"
> Responses are scored on a four-category ordinal scale:
> 0 = Not at all; 1 = Several days; 2 = More than half the days; 3 = Nearly every day.
> The PHQ-8 item stems (corresponding to variables `phq1`–`phq8`) are:
>
> 1. Little interest or pleasure in doing things
> 2. Feeling down, depressed, irritable, or hopeless
> 3. Trouble falling or staying asleep, or sleeping too much
> 4. Feeling tired or having little energy
> 5. Poor appetite or overeating
> 6. Feeling bad about yourself — or that you are a failure or have let yourself or your family down
> 7. Trouble concentrating on things, such as schoolwork, reading, or watching television
> 8. Moving or speaking so slowly that other people could have noticed, or the opposite — being so fidgety or restless that you have been moving around much more than usual
>
> These item stems define the observed indicators used in all CFA and multigroup measurement invariance models in this workflow.

Required Input Files and Project Structure

The workflow assumes the following project directory structure (relative to the Quarto document):

- `data/PHQ_FINAL_manuscript.sav` — primary analytic dataset (SPSS format)
- `data/phq_final.csv` — CSV copy generated by the workflow

- output folders generated during execution (e.g., `cfa_total/`, `cfa_2A/`, `cfa_2B/`, `invariance/`)

Data Access

The analytic dataset used in this study will be made available as a de-identified research file accompanying this supplement. A direct access link will be provided here upon publication:

> **Data repository link (placeholder):** □ *[link to be inserted]*

All scripts in this supplement reproduce the analyses exactly when run with the provided dataset and software versions documented below.

Decision Log — Modeling and Reproducibility Assumptions

Decision Area	Final Choice	Rationale
Estimator	WLSMV with polychoric correlations	Appropriate for ordinal indicators and categorical response structures under THETA parameterization
Identification	Unit loading identification	Ensures comparability across subgroups
Grouping variables	Ethnicity and sex	Matches study aims and subgroup comparisons reported in manuscript
Invariance sequence	Configural → Metric → Scalar	Standard hierarchical approach for multigroup CFA
Fit evaluation	CFI, SRMR, ΔCFI, ΔSRMR	Index set appropriate for categorical indicators under WLSMV
Missing values	Prespecified missing□code recoding to NA	Minimizes bias given <5% missingness
Reproducibility	Single executable Quarto workflow	Ensures full transparency of analytic pipeline

> **Manuscript tables were typeset in a word-processed format for publication clarity. All numerical values appearing in those tables are reproduced in this supplement via gt-generated computational tables to ensure verifiable analytic correspondence.**

Analytic Roadmap — Reproducible Workflow

This supplement reproduces all analyses reported in the manuscript. Each stage below includes the code, model syntax, and outputs used to generate the corresponding manuscript tables.

1. Data import and preparation

Reproduces dataset import, missing-value recoding, analytic variable selection, and construction of composite scale scores used in descriptive and validity analyses.

> **Correspondence to manuscript:** Supports sample preparation underlying Table 1.

2. Group definitions and descriptive statistics

Defines analytic subgroups across ethnicity and sex and computes item-level descriptive statistics prior to factor modeling.

> **Correspondence to manuscript:** Descriptive summaries support contextual interpretation of subsequent CFA and invariance analyses.

3. Confirmatory Factor Analyses (CFA)

Estimates competing CFA models using ordinal indicators in *lavaan*:

- **Model 1 — One-factor model**
- **Model 2A & 2B — Alternative cognitive-affective / somatic two-factor structures**

Outputs include:

- model fit indices
- standardized factor loadings
- factor correlations
- McDonald's omega reliability estimates

> **Correspondence to manuscript:** Reproduces CFA results reported in **Tables 2–6**.

4. Measurement invariance testing

Estimates multigroup configural, metric, and scalar models across ethnicity and sex using WLSMV estimation with polychoric correlations.

Outputs include:

- subgroup fit indices
- ΔCFI and ΔSRMR comparisons
- retained invariance decisions

> **Correspondence to manuscript:** Reproduces measurement invariance results reported in **Table 7**.

5. Associations with theoretically related variables

Computes subgroup-specific correlations between PHQ-8 scores and theoretically related constructs to support extrapolation inferences.

> **Correspondence to manuscript:** Reproduces validity correlations reported in **Table 8**.

> ℹ Note
>
> **How to Use This Supplement**
> This document is intended to be read sequentially, with each section building on the outputs of the preceding stage of the workflow. All analyses can be reproduced by rendering this Quarto file (to HTML or PDF), provided that R, the required packages (including **lavaan**), and the project directory structure are preserved.
> The rendered output displays the full code and corresponding results and may be exported or shared for archival or review purposes. Exact numerical reproduction may vary slightly at the fourth decimal place across software versions; all model syntax, estimation settings, and input matrices are fully documented to ensure analytic equivalence.

1. Setup and Software Environment

This supplement was prepared in R using Quarto to document the complete analytic workflow underlying the psychometric analyses reported in the manuscript. All analyses were conducted using base R and established contributed packages for data management, psychometric evaluation, table construction, and structural equation modeling in **lavaan**.

The workflow loads the required R packages for data import, scoring, descriptive summaries, CFA estimation, measurement invariance testing, and table generation.

Load R packages for data import, scoring, and tables.

```
# Core data wrangling and I/O
library(tidyverse)   # dplyr, tidyr, purrr, readr, stringr, tibble
library(here)        # project-relative file paths
library(haven)       # read_sav()

# Psychometrics / SEM
library(lavaan)      # CFA models with ordinal indicators
library(psych)       # polychoric correlations and descriptive stats

# Tables, export, and utilities
library(gt)          # model fit and loading tables
library(glue)        # string formatting for labels / filenames

# Diagrams
```

```r
library(DiagrammeR)  # CFA path diagrams
library(DiagrammeRsvg)
library(rsvg)
library(webshot2)
library(kableExtra)
```

---

2. Data source and analytic sample

The analyses reported in this supplement are based on data collected from emerging adult college students who completed the PHQ-8 and related mental health measures as part of a broader psychosocial survey. The dataset includes item-level responses for the PHQ-8, demographic characteristics (e.g., sex, ethnicity, generation status), and auxiliary measures used in descriptive and validity analyses.

This section documents the structure and contents of the analytic dataset, including variable names, coding schemes, and response distributions. No transformations, scoring procedures, or statistical models are applied in this section; all variable preparation and scale construction are described in the subsequent section.

Import the SPSS dataset

```r
# Read SPSS dataset containing the analytic sample
phq_total <- read_sav(
  here("data", "PHQ_FINAL_manuscript.sav")
)

# Recode missing-value code 99 to NA across all variables
phq_total[phq_total == 99] <- NA
```

Sample characteristics

Table 1 summarizes core demographic characteristics of the analytic sample by ethnicity and sex. For Filipino Americans and European Americans, we report mean age and standard deviation, sex composition, academic classification (freshman–senior), and generation status. These estimates are computed directly from the analytic dataset used in all subsequent analyses and reproduce the sample characteristics presented in the manuscript.

```r
fil_total  <- phq_total %>% dplyr::filter(filipino == 1)
fil_women  <- phq_total %>% dplyr::filter(filipino == 1, female == 1)
fil_men    <- phq_total %>% dplyr::filter(filipino == 1, female == 0)

ea_total   <- phq_total %>% dplyr::filter(filipino == 0)
ea_women   <- phq_total %>% dplyr::filter(filipino == 0, female == 1)
ea_men     <- phq_total %>% dplyr::filter(filipino == 0, female == 0)
```

```r
n_fil_total <- nrow(fil_total)
n_fil_women <- nrow(fil_women)
n_fil_men   <- nrow(fil_men)

n_ea_total  <- nrow(ea_total)
n_ea_women  <- nrow(ea_women)
n_ea_men    <- nrow(ea_men)

age_summary <- function(df) {
  c(
    mean = mean(df$age, na.rm = TRUE),
    sd   = stats::sd(df$age, na.rm = TRUE)
  )
}

prop_pct <- function(x, levels_vec) {
  tab <- prop.table(table(factor(x, levels = levels_vec))) * 100
  as.numeric(tab)
}

age_fil_total <- age_summary(fil_total)
age_fil_women <- age_summary(fil_women)
age_fil_men   <- age_summary(fil_men)

age_ea_total  <- age_summary(ea_total)
age_ea_women  <- age_summary(ea_women)
age_ea_men    <- age_summary(ea_men)

age_df <- dplyr::bind_rows(
  tibble::tibble(
    Domain   = "Age (18-25 years)",
    Category = "M",
    Fil_Total  = age_fil_total["mean"],
    Fil_Women  = age_fil_women["mean"],
    Fil_Men    = age_fil_men["mean"],
    EA_Total   = age_ea_total["mean"],
    EA_Women   = age_ea_women["mean"],
    EA_Men     = age_ea_men["mean"]
  ),
  tibble::tibble(
    Domain   = "Age (18-25 years)",
    Category = "SD",
    Fil_Total  = age_fil_total["sd"],
    Fil_Women  = age_fil_women["sd"],
    Fil_Men    = age_fil_men["sd"],
    EA_Total   = age_ea_total["sd"],
    EA_Women   = age_ea_women["sd"],
    EA_Men     = age_ea_men["sd"]
```

```r
  )
)

sex_levels <- c(0, 1)

sex_fil <- prop_pct(fil_total$female, levels_vec = sex_levels)
sex_ea  <- prop_pct(ea_total$female,  levels_vec = sex_levels)

sex_df <- dplyr::bind_rows(
  tibble::tibble(
    Domain   = "sex",
    Category = "Men",
    Fil_Total  = sex_fil[1],
    Fil_Women  = NA_real_,
    Fil_Men    = NA_real_,
    EA_Total   = sex_ea[1],
    EA_Women   = NA_real_,
    EA_Men     = NA_real_
  ),
  tibble::tibble(
    Domain   = "sex",
    Category = "Women",
    Fil_Total  = sex_fil[2],
    Fil_Women  = NA_real_,
    Fil_Men    = NA_real_,
    EA_Total   = sex_ea[2],
    EA_Women   = NA_real_,
    EA_Men     = NA_real_
  )
)

classification_levels <- 1:4
classification_labels <- c("Freshmen", "Sophomores", "Juniors", "Seniors")

class_fil_total <- prop_pct(fil_total$classification, levels_vec =
↪  classification_levels)
class_fil_women <- prop_pct(fil_women$classification, levels_vec =
↪  classification_levels)
class_fil_men   <- prop_pct(fil_men$classification,   levels_vec =
↪  classification_levels)

class_ea_total  <- prop_pct(ea_total$classification,  levels_vec =
↪  classification_levels)
class_ea_women  <- prop_pct(ea_women$classification,  levels_vec =
↪  classification_levels)
class_ea_men    <- prop_pct(ea_men$classification,    levels_vec =
↪  classification_levels)
```

```
class_df <- purrr::map_dfr(
  seq_along(classification_levels),
  ~ tibble::tibble(
    Domain   = "Classifications",
    Category = classification_labels[.x],
    Fil_Total  = class_fil_total[.x],
    Fil_Women  = class_fil_women[.x],
    Fil_Men    = class_fil_men[.x],
    EA_Total   = class_ea_total[.x],
    EA_Women   = class_ea_women[.x],
    EA_Men     = class_ea_men[.x]
  )
)

gen_levels <- 1:3
gen_labels <- c("1st generation", "2nd generation", "3rd generation")

gen_fil_total <- prop_pct(fil_total$genstat, levels_vec = gen_levels)
gen_fil_women <- prop_pct(fil_women$genstat, levels_vec = gen_levels)
gen_fil_men   <- prop_pct(fil_men$genstat,   levels_vec = gen_levels)

gen_ea_total <- prop_pct(ea_total$genstat, levels_vec = gen_levels)
gen_ea_women <- prop_pct(ea_women$genstat, levels_vec = gen_levels)
gen_ea_men   <- prop_pct(ea_men$genstat,   levels_vec = gen_levels)

gen_df <- purrr::map_dfr(
  seq_along(gen_levels),
  ~ {
    fil_total_val <- gen_fil_total[.x]
    fil_women_val <- gen_fil_women[.x]
    fil_men_val   <- gen_fil_men[.x]

    if (gen_labels[.x] == "3rd generation") {
      ea_total_val <- gen_ea_total[.x]
      ea_women_val <- gen_ea_women[.x]
      ea_men_val   <- gen_ea_men[.x]
    } else {
      ea_total_val <- NA_real_
      ea_women_val <- NA_real_
      ea_men_val   <- NA_real_
    }

    tibble::tibble(
      Domain   = "Generation statuses",
      Category = gen_labels[.x],
      Fil_Total  = fil_total_val,
      Fil_Women  = fil_women_val,
      Fil_Men    = fil_men_val,
```

```r
      EA_Total   = ea_total_val,
      EA_Women   = ea_women_val,
      EA_Men     = ea_men_val
    )
  }
)

demo_df <- dplyr::bind_rows(
  age_df,
  sex_df,
  class_df,
  gen_df
)
```

Construct Demographic Sample Characteristics Table

```r
sample_table_gt <- demo_df %>%
  gt::gt(
    rowname_col   = "Category",
    groupname_col = "Domain"
  ) %>%
  gt::cols_label(
    Fil_Total = gt::md(paste0("Filipino Americans<br>(n = ", n_fil_total, ")")),
    Fil_Women = gt::md(paste0("Filipino American Women<br>(n = ", n_fil_women,
     ↪  ")")),
    Fil_Men   = gt::md(paste0("Filipino American Men<br>(n = ", n_fil_men, ")")),
    EA_Total  = gt::md(paste0("European Americans<br>(n = ", n_ea_total, ")")),
    EA_Women  = gt::md(paste0("European American Women<br>(n = ", n_ea_women,
     ↪  ")")),
    EA_Men    = gt::md(paste0("European American Men<br>(n = ", n_ea_men, ")"))
  ) %>%
  gt::fmt_number(
    rows    = Domain == "Age (18-25 years)",
    columns = dplyr::all_of(c("Fil_Total","Fil_Women","Fil_Men",
                              "EA_Total","EA_Women","EA_Men")),
    decimals = 2
  ) %>%
  gt::fmt_number(
    rows    = Domain != "Age (18-25 years)",
    columns = dplyr::all_of(c("Fil_Total","Fil_Women","Fil_Men",
                              "EA_Total","EA_Women","EA_Men")),
    decimals = 1,
    pattern  = "{x}%"
  ) %>%
  gt::fmt_missing(
    columns = dplyr::everything(),
    missing_text = "-"
```

```
  ) %>%
  gt::tab_header(
    title = "Demographic Characteristics of the Sample"
  )
```

Render the sample demographic characteristics table for HTML output and save a high-resolution PNG version for use in the PDF.

```
sample_table_gt

  gt::gtsave(
    data     = sample_table_gt,
    filename = "Table1_Demographic_Characteristics.png",
    path     = "tables",
    zoom     = 2
  )
```

Insert the saved demographic characteristics table image into the PDF version of the supplement.

```
knitr::include_graphics("tables/Table1_Demographic_Characteristics.png")
```

Demographic Characteristics of the Sample

	Filipino Americans (n = 698)	Filipino American Women (n = 419)	Filipino American Men (n = 279)	European Americans (n = 945)	European American Women (n = 559)	European American Men (n = 386)
Age (18–25 years)						
M	19.16	19.08	19.28	19.51	19.50	19.53
SD	1.46	1.39	1.56	1.71	1.72	1.69
sex						
Men	40.0%	—	—	40.8%	—	—
Women	60.0%	—	—	59.2%	—	—
Classifications						
Freshmen	54.7%	54.7%	54.8%	47.7%	48.8%	46.1%
Sophomores	23.4%	23.4%	23.3%	27.3%	24.2%	31.9%
Juniors	14.6%	13.8%	15.8%	15.7%	17.4%	13.2%
Seniors	7.3%	8.1%	6.1%	9.3%	9.7%	8.8%
Generation statuses						
1st generation	37.3%	41.0%	32.1%	—	—	—
2nd generation	61.0%	57.7%	65.7%	—	—	—
3rd generation	1.7%	1.3%	2.3%	100.0%	100.0%	100.0%

3. Variable preparation and scale construction

This section documents all data preparation steps required prior to model estimation. Item-level variables are recoded to ensure consistent directionality, missing values are handled using prespecified rules, and composite scale scores are computed for use in descriptive, correlational, and validity analyses.

Specifically, reverse-coded CES-D items are transformed, and mean scores are calculated for the PHQ-8, generalized anxiety, perceived stress, and CES-D depressive symptom measures. Additional composite variables are constructed for brooding rumination, self-deprecation and positive esteem (derived from the Rosenberg Self-Esteem Scale), and overall quality of life. At the conclusion of this section, the dataset contains a finalized set of analytic variables that are used unchanged in all subsequent subgroup summaries, confirmatory factor analyses, measurement invariance models, and correlational validity analyses.

Scale construction and composite variable creation

```
# Recode CES-D items

phq_total$cesd12r <- 3 - phq_total$cesd12
phq_total$cesd16r <- 3 - phq_total$cesd16

# Scale scores

phq_total$phq_mean <- rowMeans(subset(phq_total, select = c(phq1:phq8)), na.rm =
 ↪  TRUE)

phq_total$gad_mean <- rowMeans(subset(phq_total, select = c(gad1:gad7)), na.rm =
 ↪  TRUE)

phq_total$stress_mean <- rowMeans(subset(phq_total, select =
 ↪  c(stress1:stress10)), na.rm = TRUE)

phq_total$cesd_mean <- rowMeans(
subset(
phq_total,
select = c(cesd6:cesd11, cesd14:cesd15, cesd18:cesd20, cesd12r, cesd16r)
),
na.rm = TRUE
)

# Brooding rumination: rum1 rum3 rum6 rum7 rum8

phq_total$brood_mean <- rowMeans(
subset(phq_total, select = c(rum1, rum3, rum6, rum7, rum8)),
na.rm = TRUE
)

# Self-deprecation: Esteem1 Esteem2 Esteem5 Esteem9 Esteem10

phq_total$selfdep_mean <- rowMeans(
```

```
subset(phq_total, select = c(Esteem1, Esteem2, Esteem5, Esteem9, Esteem10)),
na.rm = TRUE
)

# Positive esteem: Esteem3 Esteem4 Esteem6 Esteem7 Esteem8

phq_total$posesteem_mean <- rowMeans(
subset(phq_total, select = c(Esteem3, Esteem4, Esteem6, Esteem7, Esteem8)),
na.rm = TRUE
)

# Keep only analysis variables needed downstream

phq_total <- phq_total %>%
select(
classification,
female,
age,
filipino,
genstat,
phq1:phq8,
phq_mean,
gad_mean,
stress_mean,
cesd_mean,
brood_mean,
selfdep_mean,
posesteem_mean,
qol
)
```

---

4. Group definitions and descriptive statistics

This section implements the subgroup definitions and descriptive procedures outlined in the analytic roadmap and applies them within the executable workflow. Building on the analytic roadmap, this section implements the subgroup definitions used throughout the confirmatory factor and measurement invariance analyses and presents item-level descriptive statistics for each group. Participants are classified based on ethnicity and sex to support comparisons between Filipino American and European American respondents, as well as between men and women within each population.

Descriptive statistics for PHQ-8 items (means and standard deviations) are reported separately for each subgroup to characterize observed response distributions prior to factor modeling. These summaries provide empirical context for subsequent analyses and should be interpreted as descriptive rather than inferential.

Item-level means and standard deviations summarize observed response patterns and are reported to reproduce the descriptive values referenced in the manuscript tables. Polychoric correlations used for confirmatory factor analysis are reported separately to reflect the association structure used in model estimation with ordinal indicators.

Analytic subgroups and variable labeling

```
# Create ethnicity-by-sex grouping indicators and total-group flags
phq_total <- phq_total %>%
  mutate(
    Fil_Women   = ifelse(filipino == 1 & female == 1, 1, NA),
    Fil_Men     = ifelse(filipino == 1 & female == 0, 1, NA),
    EA_Women    = ifelse(filipino == 0 & female == 1, 1, NA),
    EA_Men      = ifelse(filipino == 0 & female == 0, 1, NA),

    # Ethnicity-only group indicators
    Fil_Amer    = ifelse(filipino == 1, 1, NA),
    Eur_Amer    = ifelse(filipino == 0, 1, NA),

    # Overall sex totals
    Total_Men   = ifelse(female == 0, 1, NA),
    Total_Women = ifelse(female == 1, 1, NA)
  )

# Working dataset for subgroup-specific analyses
phq_subs <- phq_total

# -------------------------------------------------------------------
# Factor labels for reporting
# -------------------------------------------------------------------

female_labels         <- c("Male", "Female")
classification_labels <- c("Freshman", "Sophomore", "Junior", "Senior")
genstat_labels        <- c("1st Generation", "2nd Generation", "3rd Generation")

# Apply human-readable factor labels
phq_subs$female         <- factor(phq_subs$female, levels = c(0, 1), labels =
↪  female_labels)
phq_subs$classification <- factor(phq_subs$classification, levels = 1:4, labels
↪  = classification_labels)
phq_subs$genstat        <- factor(phq_subs$genstat, levels = 1:3, labels =
↪  genstat_labels)

# -------------------------------------------------------------------
# Define subgroup-specific PHQ-8 item datasets
# (used consistently across descriptives, correlations, and CFA models)
# -------------------------------------------------------------------

# Filipino American women / men
```

```r
subset_fw <- phq_subs %>%
  filter(Fil_Women == 1) %>%
  select(phq1:phq8)

subset_fm <- phq_subs %>%
  filter(Fil_Men == 1) %>%
  select(phq1:phq8)

# European American women / men
subset_ew <- phq_subs %>%
  filter(EA_Women == 1) %>%
  select(phq1:phq8)

subset_em <- phq_subs %>%
  filter(EA_Men == 1) %>%
  select(phq1:phq8)

# Ethnicity totals
subset_fa <- phq_subs %>%
  filter(Fil_Amer == 1) %>%
  select(phq1:phq8)

subset_ea <- phq_subs %>%
  filter(Eur_Amer == 1) %>%
  select(phq1:phq8)

# Overall men / women totals
subset_total_men <- phq_subs %>%
  filter(Total_Men == 1) %>%
  select(phq1:phq8)

subset_total_women <- phq_subs %>%
  filter(Total_Women == 1) %>%
  select(phq1:phq8)
```

Item-level descriptive statistics by subgroup

Item-level means and standard deviations are computed from the same subgroup-specific item responses used for CFA estimation and are reported alongside the polychoric correlation matrices extracted from the lavaan estimation inputs. Descriptive statistics are therefore presented within the correlation tables rather than as a separate standalone table.

```r
# Function to compute item-level means and standard deviations
calculate_stats <- function(data) {
  means <- sapply(data, mean, na.rm = TRUE)
  sds   <- sapply(data, sd,   na.rm = TRUE)

```

```
  list(
    means = means,
    sds   = sds
  )
}

# -------------------------------------------------------------------
# Compute descriptive statistics for each analytic subgroup
# -------------------------------------------------------------------

stats_fw         <- calculate_stats(subset_fw)
stats_fm         <- calculate_stats(subset_fm)
stats_ew         <- calculate_stats(subset_ew)
stats_em         <- calculate_stats(subset_em)
stats_fa         <- calculate_stats(subset_fa)
stats_ea         <- calculate_stats(subset_ea)
stats_total_men  <- calculate_stats(subset_total_men)
stats_total_women<- calculate_stats(subset_total_women)

# -------------------------------------------------------------------
# Store subgroup descriptive statistics in a named list
# (used later when constructing combined correlation-descriptive tables)
# -------------------------------------------------------------------

all_stats <- list(
  Filipino_Women      = stats_fw,
  Filipino_Men        = stats_fm,
  EA_Women            = stats_ew,
  EA_Men              = stats_em,
  Filipino_Americans  = stats_fa,
  European_Americans  = stats_ea,
  Total_Men           = stats_total_men,
  Total_Women         = stats_total_women
)
```

---

5. Confirmatory factor analytic models

> **Correspondence to manuscript.** Outputs in this section reproduce the CFA model fit indices, standardized factor loadings, factor correlations, and McDonald's omega reliability estimates reported in Tables 2–6 of the manuscript. CFA models in this section are estimated using polychoric correlations for ordinal indicators under WLSMV estimation.

A series of confirmatory factor analytic (CFA) models were estimated to evaluate competing latent structures of the PHQ-8 using ordinal item indicators. All models were estimated separately across analytic subgroups to assess the stability of the factor structure across ethnicity and sex. Model fit indices, standardized factor loadings, and factor correlations are reported for each specification.

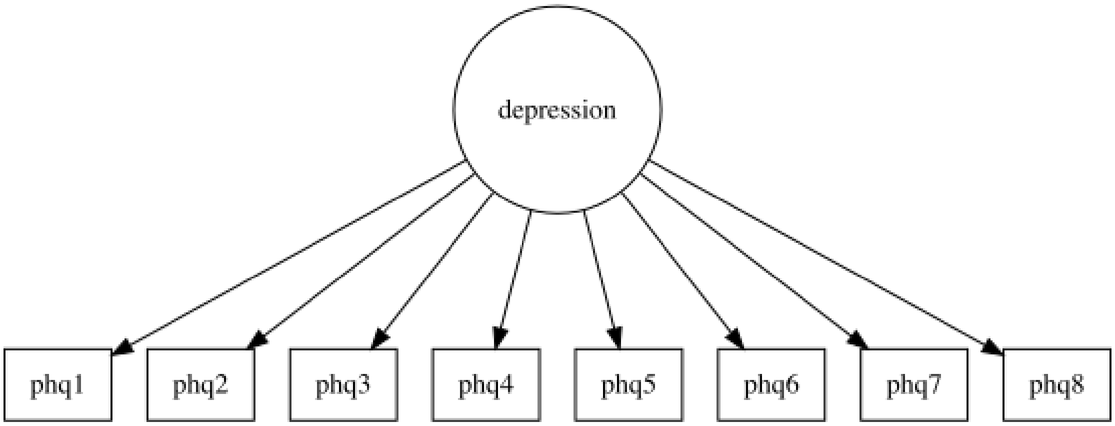
5.1 One-factor model

The one-factor model specifies a single latent depression factor underlying all eight PHQ-8 items. This model serves as the baseline representation of PHQ-8 dimensionality and provides a reference point for evaluating alternative factor structures. Model fit statistics, standardized factor loadings, and item correlations are examined across all analytic subgroups.

Diagram of the one-factor PHQ-8 measurement model.

```
# ------------------------------------------------------------------
# Diagram code for one-factor PHQ-8 model
# ------------------------------------------------------------------

diagram_code_onefactor <- "
digraph {

  # Graph properties
  label = 'One Factor Model of the PHQ-8'
  graph [ranksep = .3]

  # Observed indicators
  node [shape = box]
  phq1;
  phq2;
  phq3;
  phq4;
  phq5;
  phq6;
  phq7;
  phq8;

  # Latent factor
  node [shape = circle]
  depression;

  # Factor loadings
  edge [minlen = 3.5]
  depression -> phq1
  depression -> phq2
  depression -> phq3
  depression -> phq4
  depression -> phq5
```

```
  depression -> phq6
  depression -> phq7
  depression -> phq8
}
"

# Build DiagrammeR graph object
onefactor_graph <- DiagrammeR::grViz(diagram_code_onefactor)

# Ensure static PNG exists for PDF output
svg_txt <- DiagrammeRsvg::export_svg(onefactor_graph)
rsvg::rsvg_png(
  charToRaw(svg_txt),
  file = "figures/cfa_onefactor_phq8.png"
)

# -------------------------------------------------------------------
# Format-specific rendering
# -------------------------------------------------------------------

if (knitr::is_html_output()) {

  # HTML / IDE preview: interactive DiagrammeR widget
  onefactor_graph

} else if (knitr::is_latex_output()) {

  # PDF (and other LaTeX-based outputs): use static image
  knitr::include_graphics("figures/cfa_onefactor_phq8.png")

} else {

  # Fallback for other formats
  onefactor_graph
}
```



One Factor Model of the PHQ-8

One-factor CFA model (estimated separately by subgroup)

ℹ Syntax scaffolding: One-factor CFA models in lavaan

- **Goal.** This chunk estimates a **single-factor CFA model** for the PHQ-8 and reproduces the fit indices and loadings for the one-factor moedl reported in Tables 2–4.
- **Model specification (lavaan).**
  The object `phq_cfa_model` is a character string in lavaan model syntax defining one latent factor (`dep`) with indicators `phq1`–`phq8`, written in the form `dep =~ phq1 + phq2 + … + phq8`. This syntax is passed directly to `lavaan::cfa()`.
- This syntax is passed directly to `lavaan::cfa()`.
  - **Estimator and parameterization.**
    The helper function `run_cfa()` calls **`lavaan::cfa()`** with:
    * `ordered = c("phq1", ..., "phq8")` so that all PHQ items are treated as **ordinal** indicators.
    * `estimator = "WLSMV"` to match the manuscript (DWLS with robust corrections; recommended for ordinal items).
    * `parameterization = "theta"` to use the **THETA parameterization**, which expresses residual variances directly and aligns with the LTA workflow.
  - **Subgroup estimation.**
    The calls
  - fit_total <- run_cfa(…) fit_total_men <- run_cfa(subset_total_men) fit_total_women <- run_cfa(subset_total_women)
  - run the *same* CFA model separately for each analytic subgroup (total sample, Filipino Americans, European Americans, and sex-specific subsamples).
  - **Fit indices.**
    The object `CFA_summary` is built by calling **`lavaan::fitMeasures()`** on each fitted object (e.g., `fit_total`, `fit_fa`), extracting:

    * Number of parameters ("npar"), χ^2 and df ("chisq", "df"), p-value ("pvalue"),
    * Global fit indices (e.g., "cfi", "tli", "rmsea", "srmr").
      These are then assembled into a data frame and formatted using `gt::gt()` for reporting.
  - **Software dependencies.**
    * **lavaan**: `cfa()`, `fitMeasures()`
    * **dplyr** (via %>%), **stats** (base R functions), and **gt** for table formatting.

```
# ------------------------------------------------------------------
# One-factor PHQ-8 CFA model (ordinal indicators; WLSMV estimator, THETA)
# ------------------------------------------------------------------

phq_cfa_model <- "
  dep =~ phq1 + phq2 + phq3 + phq4 +
         phq5 + phq6 + phq7 + phq8
"

run_cfa <- function(data) {
  cfa(
    model           = phq_cfa_model,
    data            = data,
    ordered         = c('phq1','phq2','phq3','phq4',
                        'phq5','phq6','phq7','phq8'),
    estimator       = "WLSMV",
    parameterization = "theta"
  )
}

# Total analytic sample
fit_total <- run_cfa(
  phq_subs %>%
    select(phq1:phq8)
)

# Subgroup models
fit_total_men   <- run_cfa(subset_total_men)
fit_total_women <- run_cfa(subset_total_women)

fit_fm <- run_cfa(subset_fm)
fit_fw <- run_cfa(subset_fw)

fit_em <- run_cfa(subset_em)
fit_ew <- run_cfa(subset_ew)

fit_fa <- run_cfa(subset_fa)
fit_ea <- run_cfa(subset_ea)
```

Model fit indices for the one-factor PHQ-8 model across subgroups.

Fit index reporting policy. All fit indices reproduced in this supplement are extracted directly from the lavaan `fitMeasures()` output for models estimated using WLSMV with ordered indicators under THETA parameterization.

```
# Collect one-factor CFA fit indices across analytic subgroups
CFA_summary <- data.frame(
  Model = c(
    "Total Sample",
    "Total Men",
    "Total Women",
    "Filipino Americans",
    "Filipino American Men",
    "Filipino American Women",
    "European Americans",
    "European American Men",
    "European American Women"
  ),
  Par = c(
    fitMeasures(fit_total,       "npar"),
    fitMeasures(fit_total_men,   "npar"),
    fitMeasures(fit_total_women, "npar"),
    fitMeasures(fit_fa,          "npar"),
    fitMeasures(fit_fm,          "npar"),
    fitMeasures(fit_fw,          "npar"),
    fitMeasures(fit_ea,          "npar"),
    fitMeasures(fit_em,          "npar"),
    fitMeasures(fit_ew,          "npar")
  ),
  ChiSq = c(
    fitMeasures(fit_total,       "chisq"),
    fitMeasures(fit_total_men,   "chisq"),
    fitMeasures(fit_total_women, "chisq"),
    fitMeasures(fit_fa,          "chisq"),
    fitMeasures(fit_fm,          "chisq"),
    fitMeasures(fit_fw,          "chisq"),
    fitMeasures(fit_ea,          "chisq"),
    fitMeasures(fit_em,          "chisq"),
    fitMeasures(fit_ew,          "chisq")
  ),
  df = c(
    fitMeasures(fit_total,       "df"),
    fitMeasures(fit_total_men,   "df"),
    fitMeasures(fit_total_women, "df"),
    fitMeasures(fit_fa,          "df"),
```

```
    fitMeasures(fit_fm,          "df"),
    fitMeasures(fit_fw,          "df"),
    fitMeasures(fit_ea,          "df"),
    fitMeasures(fit_em,          "df"),
    fitMeasures(fit_ew,          "df")
  ),
  p_value = c(
    fitMeasures(fit_total,       "pvalue"),
    fitMeasures(fit_total_men,   "pvalue"),
    fitMeasures(fit_total_women, "pvalue"),
    fitMeasures(fit_fa,          "pvalue"),
    fitMeasures(fit_fm,          "pvalue"),
    fitMeasures(fit_fw,          "pvalue"),
    fitMeasures(fit_ea,          "pvalue"),
    fitMeasures(fit_em,          "pvalue"),
    fitMeasures(fit_ew,          "pvalue")
  ),
  CFI = c(
    fitMeasures(fit_total,       "cfi"),
    fitMeasures(fit_total_men,   "cfi"),
    fitMeasures(fit_total_women, "cfi"),
    fitMeasures(fit_fa,          "cfi"),
    fitMeasures(fit_fm,          "cfi"),
    fitMeasures(fit_fw,          "cfi"),
    fitMeasures(fit_ea,          "cfi"),
    fitMeasures(fit_em,          "cfi"),
    fitMeasures(fit_ew,          "cfi")
  ),
  SRMR = c(
    fitMeasures(fit_total,       "srmr"),
    fitMeasures(fit_total_men,   "srmr"),
    fitMeasures(fit_total_women, "srmr"),
    fitMeasures(fit_fa,          "srmr"),
    fitMeasures(fit_fm,          "srmr"),
    fitMeasures(fit_fw,          "srmr"),
    fitMeasures(fit_ea,          "srmr"),
    fitMeasures(fit_em,          "srmr"),
    fitMeasures(fit_ew,          "srmr")
  ),
  stringsAsFactors = FALSE
)

# Format fit statistics for reporting
CFA_summary_print <-
  CFA_summary %>%
  dplyr::mutate(
    ChiSq  = round(ChiSq, 2),
    CFI    = round(CFI, 3),
```

```
    SRMR    = round(SRMR, 3),
    p_value = ifelse(
      p_value < 0.001,
      "<.001",
      sprintf("%.3f", p_value)
    )
  )

# Build gt table object
CFA_summary_table <-
  CFA_summary_print %>%
  gt::gt() %>%
  gt::cols_label(
    Model    = "Model",
    Par      = "Par",
    ChiSq    = "ChiSq",
    df       = gt::md("*df*"),
    p_value  = gt::md("*p-value*"),
    CFI      = "CFI",
    SRMR     = "SRMR"
  ) %>%
  gt::tab_header(
    title = "Fit Statistics for the One-Factor PHQ-8 Model"
  )
```

Note on correspondence to manuscript tables.
The fit indices reported in this table reproduce the numerical values used in the manuscript CFA results. The table formatting here is optimized for reproducibility and inspection of the analytic workflow, rather than for publication layout. The manuscript versions of these tables present the same values in a condensed, publication-formatted structure.

Render the one-factor PHQ-8 CFA model fit indices for all analytic groups in HTML, and export a high-resolution PNG version for inclusion in the PDF.

```
CFA_summary_table

gt::gtsave(
  data     = CFA_summary_table,
  filename = "table4_onefactor_cfa_fit.png",
  path     = "tables",
  zoom     = 2
)
```

Insert the saved one-factor PHQ-8 CFA fit table image into the PDF version of the supplement.

```
knitr::include_graphics("tables/table4_onefactor_cfa_fit.png")
```

Fit Statistics for the One-Factor PHQ-8 Model						
Model	Par	ChiSq	*df*	*p-value*	CFI	SRMR
Total Sample	32	264.06	20	<.001	0.991	0.050
Total Men	32	115.37	20	<.001	0.990	0.058
Total Women	32	180.97	20	<.001	0.991	0.053
Filipino Americans	32	149.08	20	<.001	0.989	0.058
Filipino American Men	32	68.59	20	<.001	0.989	0.066
Filipino American Women	32	111.94	20	<.001	0.987	0.065
European Americans	32	129.29	20	<.001	0.993	0.048
European American Men	32	64.29	20	<.001	0.991	0.060
European American Women	32	84.87	20	<.001	0.995	0.048

Extract PHQ-8 item correlation matrices

Correlations are computed as polychoric association matrices to reflect the correlation structure used in CFA estimation with ordinal indicators in lavaan. These matrices are presented by subgroup to document the correlation inputs underlying model estimation and to support reproducibility of the correlations reported in the manuscript.

Note on descriptive summaries and model inputs. Item means and standard deviations are reported within the same tables as the subgroup polychoric correlations to maintain fidelity to the model input matrices used in estimation.

```
# ------------------------------------------------------------------
# Define PHQ-8 indicator set used in CFA estimation
# ------------------------------------------------------------------

phq_items <- paste0("phq", 1:8)

# Assemble subgroup-specific PHQ-8 item matrices
correlations_list <- list(
  Filipino_Americans = subset_fa[, phq_items],
  European_Americans = subset_ea[, phq_items],
  Total_Men          = subset_total_men[, phq_items],
  Total_Women        = subset_total_women[, phq_items],
  Filipino_Men       = subset_fm[, phq_items],
```

```r
  Filipino_Women     = subset_fw[, phq_items],
  EA_Men             = subset_em[, phq_items],
  EA_Women           = subset_ew[, phq_items]
)

# -------------------------------------------------------------------
# Compute polychoric correlations for each analytic subgroup
# (matches the association structure used in WLSMV CFA estimation)
# -------------------------------------------------------------------

correlations_list <- lapply(
  correlations_list,
  function(df) psych::polychoric(df)$rho
)
```

PHQ-8 polychoric correlation matrices by analytic subgroup.

Define subgroup-specific item datasets for correlation–descriptive table construction.

This mapping links each analytic subgroup to the corresponding item-level PHQ-8 responses used to compute observed item means and standard deviations. These subgroup datasets are paired with the subgroup-specific polychoric correlation matrices used in CFA estimation to produce combined correlation–descriptive tables..

```r
# Mapping of analytic subgroup names to the corresponding PHQ-8 item datasets
# (used when constructing combined correlation-descriptive tables)

subset_map <- list(
  Filipino_Americans = subset_fa,
  European_Americans = subset_ea,
  Total_Men          = subset_total_men,
  Total_Women        = subset_total_women,
  Filipino_Men       = subset_fm,
  Filipino_Women     = subset_fw,
  EA_Men             = subset_em,
  EA_Women           = subset_ew
)
```

Construct combined polychoric correlation tables with observed item means and standard deviations.

This function formats subgroup-specific PHQ-8 polychoric correlation matrices used in CFA estimation and appends observed item means and standard deviations computed from the same subgroup item responses. The resulting tables present all descriptive inputs underlying CFA estimation in a single, report-ready format.

```r
# Construct a PHQ-8 table combining polychoric correlations with item means and
↪ SDs
make_corr_desc_gt <- function(corr_mat, item_data, title_text) {

  # -----------------------------------------------------------------
  # Prepare correlation matrix (lower-triangular display only)
  # -----------------------------------------------------------------
  corr_mat <- as.matrix(corr_mat)
  colnames(corr_mat) <- toupper(colnames(corr_mat))
  rownames(corr_mat) <- toupper(rownames(corr_mat))

  corr_disp <- round(corr_mat, 2)
  corr_disp[upper.tri(corr_disp, diag = FALSE)] <- NA

  # -----------------------------------------------------------------
  # Compute item means and standard deviations
  # -----------------------------------------------------------------
  item_data <- as.data.frame(item_data)
  names(item_data) <- toupper(names(item_data))

  item_means <- round(sapply(item_data, mean, na.rm = TRUE), 2)
  item_sds   <- round(sapply(item_data, sd,   na.rm = TRUE), 2)

  # Base data frame from correlation matrix
  df <- as.data.frame(corr_disp)
  df <- tibble::rownames_to_column(df, var = "PHQ Item")

  # Build Mean / SD rows as single-row data frames with matching columns
  mean_row <- as.data.frame(as.list(item_means[colnames(corr_disp)]))
  sd_row   <- as.data.frame(as.list(item_sds[colnames(corr_disp)]))

  mean_row <- tibble::add_column(mean_row, `PHQ Item` = "Mean", .before = 1)
  sd_row   <- tibble::add_column(sd_row,   `PHQ Item` = "SD",   .before = 1)

  # Append Mean and SD rows to the correlation matrix
  df <- dplyr::bind_rows(df, mean_row, sd_row)

  # Ensure numeric columns are treated as numeric for formatting
  num_cols <- setdiff(names(df), "PHQ Item")
  df[num_cols] <- lapply(df[num_cols], as.numeric)

  # -----------------------------------------------------------------
  # Format combined correlation-descriptive table with gt
  # -----------------------------------------------------------------
  gt(df) %>%
    cols_label(`PHQ Item` = "PHQ Items") %>%

# Enforce consistent table width & alignment (older gt-compatible)
```

```r
tab_options(
  table.width = "100%"
) %>%
cols_align(
  align = "center",
  columns = everything()
) %>%

    fmt_number(columns = all_of(num_cols), decimals = 2) %>%
    fmt_missing(columns = all_of(num_cols), missing_text = "") %>%
    tab_header(title = title_text)
}
```

Generate subgroup-specific PHQ-8 correlation–descriptive tables.

For each analytic subgroup, the polychoric correlation matrix is combined with the observed item means and standard deviations to produce a complete descriptive summary aligned with the CFA estimation inputs.

```r
corr_desc_tables <- purrr::imap(
  correlations_list,
  ~ make_corr_desc_gt(
      corr_mat   = .x,
      item_data  = subset_map[[.y]],
      title_text = paste0(
        "PHQ-8 Polychoric Correlations with Item Means and SDs: ",
        .y
      )
    )
)
```

Save correlation–descriptive tables as high-resolution image files.

Each subgroup-specific table is exported as a PNG file for archival or review purposes. Files are written to the `tables/`directory at high resolution to ensure readability in PDF outputs.

```r
purrr::iwalk(corr_desc_tables, ~ {

  safe_name <- gsub("[^A-Za-z0-9_]+", "_", .y)

  gt::gtsave(
    .x,
    filename = paste0(safe_name, "_PHQ8_corr_means_sds.png"),
    path     = "tables",
    zoom     = 2
  )
})

invisible(NULL)
```

Manuscript correspondence and purpose of this table.
The polychoric correlations, means, and standard deviations reproduced here correspond to the descriptive statistics and validity associations summarized in the manuscript tables. In this supplement, the table is rendered in a code-forward format to document the exact values generated by the analytic pipeline and to support numerical reproducibility. The publication tables present these same numerical values in a typeset layout; the versions shown here may differ in appearance, but they reproduce the underlying estimates exactly.

Display correlation–descriptive tables in the HTML output.

For interpretive convenience, subgroup-specific tables are also rendered directly in the HTML output so that correlations and descriptive statistics can be reviewed inline.

```
if (knitr::is_html_output()) {

  purrr::walk(corr_desc_tables, function(tab) {

    gt_data <- tab[["_data"]]

    # skip empty tables
    if (is.null(gt_data) || nrow(gt_data) == 0) return(NULL)

    # prefer as_raw_html() when available
    if (utils::packageVersion("gt") >= "0.10.0") {
      html_out <- gt::as_raw_html(tab)
    } else {
      # fallback for older gt builds
      html_out <- as.character(gt:::as.tags.gt_tbl(tab))
    }

    # emit raw HTML into the document
    cat(html_out, "\n\n")
  })
}
```

Insert the saved polychoric correlation tables (with means and standard deviations) into the PDF version of the supplement.

```
knitr::include_graphics("tables/Filipino_Americans_PHQ8_corr_means_sds.png")
```

PHQ-8 Polychoric Correlations with Item Means and SDs: Filipino_Americans								
PHQ Items	PHQ1	PHQ2	PHQ3	PHQ4	PHQ5	PHQ6	PHQ7	PHQ8
PHQ1	1.00							
PHQ2	0.71	1.00						
PHQ3	0.51	0.50	1.00					
PHQ4	0.58	0.53	0.71	1.00				
PHQ5	0.54	0.54	0.67	0.64	1.00			
PHQ6	0.73	0.61	0.54	0.58	0.57	1.00		
PHQ7	0.51	0.48	0.54	0.55	0.60	0.63	1.00	
PHQ8	0.50	0.48	0.47	0.49	0.48	0.63	0.59	1.00
Mean	1.04	0.93	1.32	0.95	1.51	1.03	1.26	0.53
SD	0.88	0.86	1.08	1.04	0.97	1.04	1.01	0.83

```r
knitr::include_graphics("tables/European_Americans_PHQ8_corr_means_sds.png")
```

PHQ-8 Polychoric Correlations with Item Means and SDs: European_Americans								
PHQ Items	PHQ1	PHQ2	PHQ3	PHQ4	PHQ5	PHQ6	PHQ7	PHQ8
PHQ1	1.00							
PHQ2	0.76	1.00						
PHQ3	0.54	0.53	1.00					
PHQ4	0.59	0.54	0.64	1.00				
PHQ5	0.60	0.58	0.63	0.63	1.00			
PHQ6	0.75	0.70	0.50	0.55	0.57	1.00		
PHQ7	0.61	0.66	0.55	0.56	0.64	0.65	1.00	
PHQ8	0.55	0.56	0.55	0.58	0.54	0.61	0.65	1.00
Mean	0.95	0.73	1.20	0.85	1.41	0.78	1.03	0.41
SD	0.90	0.88	1.05	0.99	0.97	0.97	1.01	0.76

```r
knitr::include_graphics("tables/Total_Men_PHQ8_corr_means_sds.png")
```

PHQ-8 Polychoric Correlations with Item Means and SDs: Total_Men								
PHQ Items	PHQ1	PHQ2	PHQ3	PHQ4	PHQ5	PHQ6	PHQ7	PHQ8
PHQ1	1.00							
PHQ2	0.69	1.00						
PHQ3	0.50	0.51	1.00					
PHQ4	0.55	0.53	0.62	1.00				
PHQ5	0.50	0.47	0.57	0.54	1.00			
PHQ6	0.76	0.70	0.49	0.53	0.48	1.00		
PHQ7	0.49	0.51	0.50	0.53	0.58	0.61	1.00	
PHQ8	0.52	0.53	0.52	0.57	0.58	0.66	0.64	1.00
Mean	0.78	0.72	1.06	0.65	1.21	0.72	1.04	0.45
SD	0.79	0.83	0.99	0.88	0.93	0.89	0.99	0.79

```r
knitr::include_graphics("tables/Total_Women_PHQ8_corr_means_sds.png")
```

PHQ-8 Polychoric Correlations with Item Means and SDs: Total_Women								
PHQ Items	PHQ1	PHQ2	PHQ3	PHQ4	PHQ5	PHQ6	PHQ7	PHQ8
PHQ1	1.00							
PHQ2	0.76	1.00						
PHQ3	0.53	0.52	1.00					
PHQ4	0.57	0.52	0.69	1.00				
PHQ5	0.59	0.62	0.68	0.66	1.00			
PHQ6	0.73	0.64	0.52	0.57	0.61	1.00		
PHQ7	0.62	0.64	0.57	0.56	0.66	0.67	1.00	
PHQ8	0.54	0.53	0.51	0.54	0.48	0.61	0.62	1.00
Mean	1.13	0.88	1.38	1.06	1.62	0.99	1.19	0.47
SD	0.93	0.90	1.10	1.06	0.97	1.07	1.03	0.79

```r
knitr::include_graphics("tables/Filipino_Men_PHQ8_corr_means_sds.png")
```

PHQ-8 Polychoric Correlations with Item Means and SDs: Filipino_Men								
PHQ Items	PHQ1	PHQ2	PHQ3	PHQ4	PHQ5	PHQ6	PHQ7	PHQ8
PHQ1	1.00							
PHQ2	0.64	1.00						
PHQ3	0.47	0.53	1.00					
PHQ4	0.49	0.54	0.61	1.00				
PHQ5	0.52	0.50	0.60	0.54	1.00			
PHQ6	0.80	0.66	0.56	0.57	0.57	1.00		
PHQ7	0.45	0.38	0.49	0.55	0.65	0.63	1.00	
PHQ8	0.50	0.49	0.56	0.53	0.61	0.67	0.67	1.00
Mean	0.85	0.86	1.07	0.65	1.21	0.87	1.18	0.54
SD	0.81	0.85	0.99	0.86	0.94	0.97	1.00	0.87

```
knitr::include_graphics("tables/Filipino_Women_PHQ8_corr_means_sds.png")
```

PHQ-8 Polychoric Correlations with Item Means and SDs: Filipino_Women								
PHQ Items	PHQ1	PHQ2	PHQ3	PHQ4	PHQ5	PHQ6	PHQ7	PHQ8
PHQ1	1.00							
PHQ2	0.75	1.00						
PHQ3	0.51	0.49	1.00					
PHQ4	0.59	0.52	0.74	1.00				
PHQ5	0.52	0.58	0.68	0.66	1.00			
PHQ6	0.69	0.57	0.51	0.57	0.55	1.00		
PHQ7	0.55	0.54	0.57	0.55	0.58	0.63	1.00	
PHQ8	0.51	0.47	0.43	0.50	0.41	0.61	0.54	1.00
Mean	1.17	0.98	1.48	1.15	1.71	1.13	1.32	0.53
SD	0.91	0.86	1.10	1.10	0.95	1.07	1.02	0.81

```
knitr::include_graphics("tables/EA_Men_PHQ8_corr_means_sds.png")
```

PHQ-8 Polychoric Correlations with Item Means and SDs: EA_Men								
PHQ Items	PHQ1	PHQ2	PHQ3	PHQ4	PHQ5	PHQ6	PHQ7	PHQ8
PHQ1	1.00							
PHQ2	0.72	1.00						
PHQ3	0.52	0.52	1.00					
PHQ4	0.59	0.54	0.63	1.00				
PHQ5	0.48	0.45	0.55	0.55	1.00			
PHQ6	0.72	0.72	0.45	0.51	0.42	1.00		
PHQ7	0.52	0.58	0.51	0.52	0.54	0.58	1.00	
PHQ8	0.53	0.54	0.50	0.62	0.56	0.64	0.61	1.00
Mean	0.73	0.62	1.05	0.65	1.20	0.62	0.94	0.38
SD	0.77	0.79	0.99	0.90	0.93	0.82	0.97	0.73

```
knitr::include_graphics("tables/EA_Women_PHQ8_corr_means_sds.png")
```

PHQ-8 Polychoric Correlations with Item Means and SDs: EA_Women								
PHQ Items	PHQ1	PHQ2	PHQ3	PHQ4	PHQ5	PHQ6	PHQ7	PHQ8
PHQ1	1.00							
PHQ2	0.77	1.00						
PHQ3	0.54	0.53	1.00					
PHQ4	0.56	0.53	0.64	1.00				
PHQ5	0.64	0.65	0.67	0.65	1.00			
PHQ6	0.75	0.68	0.52	0.56	0.65	1.00		
PHQ7	0.66	0.69	0.57	0.57	0.71	0.69	1.00	
PHQ8	0.57	0.57	0.58	0.57	0.53	0.60	0.68	1.00
Mean	1.10	0.81	1.30	0.99	1.55	0.89	1.09	0.43
SD	0.95	0.93	1.09	1.03	0.98	1.05	1.03	0.78

Standardized factor loadings for the one-factor PHQ-8 model by subgroup.

Standardized factor loadings are reported to summarize the strength of association between each PHQ-8 item and the latent depression factor across analytic subgroups. Loadings are extracted from the **lavaan** standardized solution for the one-factor CFA model estimated with ordinal indicators. Consistency in loading magnitudes across groups provides evidence for the stability of the unidimensional structure.

```
# --- your existing loading + omega functions unchanged ---

get_loadings <- function(fit, group_label) {
  pe <- parameterEstimates(fit, standardized = TRUE)
  pe <- pe[pe$op == "=~" & pe$lhs == "dep", c("rhs","std.all")]
```

```r
  data.frame(
    Item    = toupper(pe$rhs),
    Group   = group_label,
    Loading = pe$std.all,
    stringsAsFactors = FALSE
  )
}

compute_omega_dep <- function(fit) {
  pe <- parameterEstimates(fit, standardized = TRUE)
  load <- pe[pe$op=="=~" & pe$lhs=="dep","std.all"]
  items <- pe[pe$op=="=~" & pe$lhs=="dep","rhs"]
  theta <- pe[pe$op=="~~" & pe$lhs %in% items & pe$lhs==pe$rhs,"std.all"]
  (sum(load)^2) / ((sum(load)^2) + sum(theta))
}

get_omega_row <- function(fit, group_label) {
  data.frame(Item="Omega", Group=group_label,
             Loading=compute_omega_dep(fit))
}

# --- collect loadings ---

loadings_long <- rbind(
  get_loadings(fit_total_men,   "Total Men"),
  get_loadings(fit_total_women, "Total Women"),
  get_loadings(fit_fm,          "Filipino American Men"),
  get_loadings(fit_fw,          "Filipino American Women"),
  get_loadings(fit_em,          "European American Men"),
  get_loadings(fit_ew,          "European American Women"),
  get_loadings(fit_fa,          "Filipino Americans"),
  get_loadings(fit_ea,          "European Americans")
)

omegas_long <- rbind(
  get_omega_row(fit_total_men,   "Total Men"),
  get_omega_row(fit_total_women, "Total Women"),
  get_omega_row(fit_fm,          "Filipino American Men"),
  get_omega_row(fit_fw,          "Filipino American Women"),
  get_omega_row(fit_em,          "European American Men"),
  get_omega_row(fit_ew,          "European American Women"),
  get_omega_row(fit_fa,          "Filipino Americans"),
  get_omega_row(fit_ea,          "European Americans")
)

loadings_wide <- dplyr::bind_rows(loadings_long, omegas_long) %>%
  dplyr::mutate(Loading = round(Loading, 2)) %>%
  tidyr::pivot_wider(
```

```r
    id_cols = Item,
    names_from = Group,
    values_from = Loading
  ) %>%
  dplyr::mutate(
    Item = factor(
      Item,
      levels = c("PHQ1","PHQ2","PHQ3","PHQ4",
                 "PHQ5","PHQ6","PHQ7","PHQ8","Omega")
    )
  ) %>%
  dplyr::arrange(Item)

# --- build gt object (shared) ---

one_factor_table_gt <- loadings_wide %>%
  gt::gt() %>%
  gt::tab_header(
    title = "Standardized Loadings and Omega for the One-Factor PHQ-8 Model"
  ) %>%
  gt::tab_spanner(
    label   = "Standardized Loadings (PHQ1-PHQ8) and Omega",
    columns = -Item
  )
```

Interpretation and manuscript correspondence.
The standardized factor loadings and McDonald's ω values reproduced here correspond to the estimates reported in Tables 5 and 6 of the manuscript. The tables in this supplement are presented in a code-forward format to support transparent replication of the underlying estimates. The manuscript tables present the same numerical results in a typeset publication format.

Render the one-factor PHQ-8 standardized factor loadings and McDonald's omega estimates for all analytic groups in HTML, and export a high-resolution PNG version for inclusion in the PDF.

```r
one_factor_table_gt

  gt::gtsave(
    data     = one_factor_table_gt,
    filename = "Table5_6_OneFactor_Loadings_Omega.png",
    path     = "tables",
    zoom     = 2
  )
```

Insert the saved one-factor PHQ-8 standardized loadings and McDonald's omega table image into the PDF version of the supplement.

```r
knitr::include_graphics("tables/Table5_6_OneFactor_Loadings_Omega.png")
```

Standardized Loadings and Omega for the One-Factor PHQ-8 Model

Item	Standardized Loadings (PHQ1–PHQ8) and Omega							
	Total Men	Total Women	Filipino American Men	Filipino American Women	European American Men	European American Women	Filipino Americans	European Americans
PHQ1	0.81	0.83	0.80	0.82	0.82	0.85	0.81	0.84
PHQ2	0.77	0.81	0.72	0.77	0.81	0.83	0.74	0.82
PHQ3	0.70	0.75	0.71	0.78	0.70	0.73	0.77	0.73
PHQ4	0.73	0.77	0.71	0.81	0.75	0.74	0.78	0.75
PHQ5	0.70	0.80	0.75	0.77	0.66	0.82	0.77	0.77
PHQ6	0.85	0.81	0.90	0.79	0.82	0.82	0.83	0.82
PHQ7	0.73	0.79	0.74	0.73	0.73	0.83	0.73	0.79
PHQ8	0.77	0.70	0.76	0.65	0.76	0.73	0.68	0.73
Omega	0.92	0.93	0.92	0.92	0.92	0.93	0.92	0.93

5.2 Two-factor model A

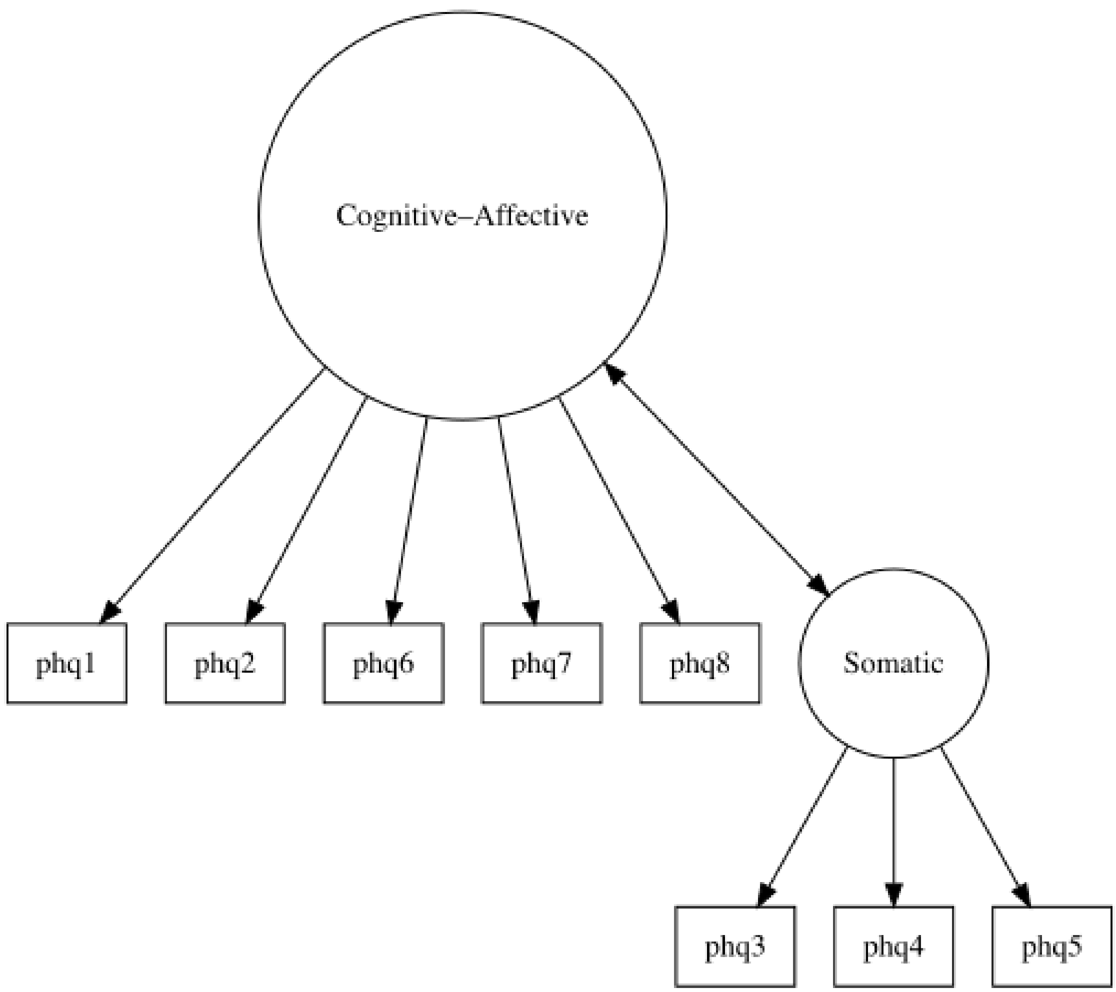
The first two-factor specification partitions PHQ-8 items into cognitive-affective and somatic symptom domains based on commonly proposed conceptual distinctions. This model evaluates whether separating cognitive-affective and somatic content improves model fit relative to the unidimensional structure. Standardized factor loadings, factor correlations, and fit indices are reported for each analytic subgroup.

In this specification, the **cognitive-affective** factor is indicated by items 3–5, and the **somatic** factor is indicated by items 1, 2, and 6–8.

```r
# ------------------------------------------------------------------
# Diagram code for Two-Factor Model A
# ------------------------------------------------------------------

diagram_code_twofactor_A <- "
digraph {

  label = 'Two Factor Model A of the PHQ-8 (Cognitive-Affective vs. Somatic
↪   factors)'
  graph [ranksep = .3]

  # Observed indicators
  node [shape = box]
  phq3;
  phq4;
```

```
  phq5;

  node [shape = box]
  phq1;
  phq2;
  phq6;
  phq7;
  phq8;

  # Latent factors
  node [shape = circle]
  Affective [label = 'Cognitive-Affective'];
  Somatic;

  # Factor loadings
  edge [minlen = 3.5]

  Affective -> phq1
  Affective -> phq2
  Affective -> phq6
  Affective -> phq7
  Affective -> phq8

  Somatic -> phq3
  Somatic -> phq4
  Somatic -> phq5

  # Correlated factors
  Affective -> Somatic [dir = both]
}
"

# Build DiagrammeR graph object
twofactor_A_graph <- DiagrammeR::grViz(diagram_code_twofactor_A)

# --------------------------------------------------------------------
# Ensure static PNG exists for PDF output
# --------------------------------------------------------------------

svg_txt <- DiagrammeRsvg::export_svg(twofactor_A_graph)
rsvg::rsvg_png(
  charToRaw(svg_txt),
  file = "figures/cfa_twofactor_modelA.png"
)

# --------------------------------------------------------------------
# Format-specific rendering
# --------------------------------------------------------------------
```

```r
if (knitr::is_html_output()) {

  # HTML / IDE preview: interactive DiagrammeR widget
  twofactor_A_graph

} else if (knitr::is_latex_output()) {

  # PDF: use static PNG
  knitr::include_graphics("figures/cfa_twofactor_modelA.png")

} else {

  # Fallback for other formats
  twofactor_A_graph
}
```



Two Factor Model A of the PHQ-8 (Cognitive–Affective vs. Somatic factors)

Two-factor Model A CFA estimated separately across analytic subgroups.

ℹ Syntax scaffolding: Two-factor PHQ-8 CFA models (Models A and B)

- **Goal.** These chunks estimate two alternative **two-factor CFA models** (Model A and Model B) that partition PHQ-8 items into cognitive-affective (AFF) and somatic (SOM) factors, and reproduce Tables 5 and 6.
- **Model specification (lavaan).**
  - Model A defines `Cognitive-Affective` and `Somatic` factors using one allocation of items (e.g., cognitive-affective: items 3–5; somatic: 1, 2, 6–8).
  - Model B defines the same two factors but with an **alternative allocation** (e.g., cognitive-affective: items 1, 2, 6–8; somatic: 3–5).
    Each specification is written as a lavaan model string (e.g., `phq_model_2A`, `phq_model_2B`) and passed into **`lavaan::cfa()`**.
- **Estimation and subgroups.**
  - As in the one-factor model, `cfa()` is called with `ordered` indicators, `estimator = "WLSMV"`, and `parameterization = "theta"`.
  - Separate fits are obtained for each analytic subgroup (total sample, Filipino Americans, European Americans, and sex-specific groups) by applying the two-factor model definitions to the subgroup data (e.g., `fit_2A_total`, `fit_2A_fa`, `fit_2B_ea`, etc.).
- **Extracting loadings and factor correlations.**
  Standardized loadings and factor correlations are extracted from each fitted object using **`lavaan::inspect()`** and related accessors, reshaped with **`dplyr`** and **`tidyr`** into long and wide formats, and then stacked across subgroups.
- **Computing McDonald's omega.**
  The helper functions `compute_omega_factor_2A()` / `compute_omega_factor_2B()` and `get_omegas_2A()` / `get_omegas_2B()` derive **McDonald's ω** for the Cognitive-Affective and Somatic factors by combining standardized loadings and residual variances from the lavaan object. The resulting omega values are:
  - Rounded with `round()`,
  - Converted to wide format with **`tidyr::pivot_wider()`**, and
  - Appended as an `"Omega"` row for each factor.
- **Tabling.**
  The combined loadings-plus-omega tables (`loadings_2A_with_omega`, `loadings_2B_with_omega`) are formatted with:
  - **`gt::gt()`** to build the table,
  - **`gt::tab_spanner()`** and `gt::tab_header()` to add a descriptive spanner and title, matching the manuscript's Tables 5–6.
- **Software dependencies.**
  - **lavaan** for CFA estimation and parameter extraction.
  - **dplyr** and **tidyr** for reshaping and combining subgroup results.
  - **gt** for publication-ready tables.

```r
# -------------------------------------------------------------------
# Two-Factor PHQ-8 CFA - Model A (WLSMV, THETA)
#   AFF = Cognitive-Affective items
#   SOM = Somatic items
# -------------------------------------------------------------------

phq_cfa_2A_model <- "
  AFF =~ phq1 + phq2 + phq6 + phq7 + phq8
  SOM =~ phq3 + phq4 + phq5
"

run_cfa_2A <- function(data) {
  cfa(
    model           = phq_cfa_2A_model,
    data            = data,
    ordered         = c('phq1','phq2','phq3','phq4',
                        'phq5','phq6','phq7','phq8'),
    estimator       = "WLSMV",
    parameterization = "theta"
  )
}

# Total analytic sample (Model A)
fit_2A_total <- run_cfa_2A(
  phq_subs %>%
    select(phq1:phq8)
)

# Subgroup models
fit_2A_total_men   <- run_cfa_2A(subset_total_men)
fit_2A_total_women <- run_cfa_2A(subset_total_women)

fit_2A_fm <- run_cfa_2A(subset_fm)
fit_2A_fw <- run_cfa_2A(subset_fw)

fit_2A_em <- run_cfa_2A(subset_em)
fit_2A_ew <- run_cfa_2A(subset_ew)

fit_2A_fa <- run_cfa_2A(subset_fa)
fit_2A_ea <- run_cfa_2A(subset_ea)
```

Model fit indices for Two-factor Model A across subgroups.

```r
# -------------------------------------------------------------------
# Collect Two-Factor Model A fit indices across analytic subgroups
# -------------------------------------------------------------------
```

```r
CFA_2A_summary <- data.frame(
  Model = c(
    "Total Sample",
    "Total Men",
    "Total Women",
    "Filipino Americans",
    "Filipino American Men",
    "Filipino American Women",
    "European Americans",
    "European American Men",
    "European American Women"
  ),
  Par = c(
    fitMeasures(fit_2A_total,       "npar"),
    fitMeasures(fit_2A_total_men,   "npar"),
    fitMeasures(fit_2A_total_women, "npar"),
    fitMeasures(fit_2A_fa,          "npar"),
    fitMeasures(fit_2A_fm,          "npar"),
    fitMeasures(fit_2A_fw,          "npar"),
    fitMeasures(fit_2A_ea,          "npar"),
    fitMeasures(fit_2A_em,          "npar"),
    fitMeasures(fit_2A_ew,          "npar")
  ),
  ChiSq = c(
    fitMeasures(fit_2A_total,       "chisq"),
    fitMeasures(fit_2A_total_men,   "chisq"),
    fitMeasures(fit_2A_total_women, "chisq"),
    fitMeasures(fit_2A_fa,          "chisq"),
    fitMeasures(fit_2A_fm,          "chisq"),
    fitMeasures(fit_2A_fw,          "chisq"),
    fitMeasures(fit_2A_ea,          "chisq"),
    fitMeasures(fit_2A_em,          "chisq"),
    fitMeasures(fit_2A_ew,          "chisq")
  ),
  df = c(
    fitMeasures(fit_2A_total,       "df"),
    fitMeasures(fit_2A_total_men,   "df"),
    fitMeasures(fit_2A_total_women, "df"),
    fitMeasures(fit_2A_fa,          "df"),
    fitMeasures(fit_2A_fm,          "df"),
    fitMeasures(fit_2A_fw,          "df"),
    fitMeasures(fit_2A_ea,          "df"),
    fitMeasures(fit_2A_em,          "df"),
    fitMeasures(fit_2A_ew,          "df")
  ),
  p_value = c(
    fitMeasures(fit_2A_total,       "pvalue"),
    fitMeasures(fit_2A_total_men,   "pvalue"),
```

```
    fitMeasures(fit_2A_total_women, "pvalue"),
    fitMeasures(fit_2A_fa,          "pvalue"),
    fitMeasures(fit_2A_fm,          "pvalue"),
    fitMeasures(fit_2A_fw,          "pvalue"),
    fitMeasures(fit_2A_ea,          "pvalue"),
    fitMeasures(fit_2A_em,          "pvalue"),
    fitMeasures(fit_2A_ew,          "pvalue")
  ),
  CFI = c(
    fitMeasures(fit_2A_total,       "cfi"),
    fitMeasures(fit_2A_total_men,   "cfi"),
    fitMeasures(fit_2A_total_women, "cfi"),
    fitMeasures(fit_2A_fa,          "cfi"),
    fitMeasures(fit_2A_fm,          "cfi"),
    fitMeasures(fit_2A_fw,          "cfi"),
    fitMeasures(fit_2A_ea,          "cfi"),
    fitMeasures(fit_2A_em,          "cfi"),
    fitMeasures(fit_2A_ew,          "cfi")
  ),
  SRMR = c(
    fitMeasures(fit_2A_total,       "srmr"),
    fitMeasures(fit_2A_total_men,   "srmr"),
    fitMeasures(fit_2A_total_women, "srmr"),
    fitMeasures(fit_2A_fa,          "srmr"),
    fitMeasures(fit_2A_fm,          "srmr"),
    fitMeasures(fit_2A_fw,          "srmr"),
    fitMeasures(fit_2A_ea,          "srmr"),
    fitMeasures(fit_2A_em,          "srmr"),
    fitMeasures(fit_2A_ew,          "srmr")
  ),
  stringsAsFactors = FALSE
)

# ---------------------------------------------------------------------
# Apply rounding and p-value formatting once (shared by HTML / PDF)
# ---------------------------------------------------------------------

CFA_2A_summary_print <-
  CFA_2A_summary %>%
  dplyr::mutate(
    ChiSq   = round(ChiSq, 2),
    CFI     = round(CFI, 3),
    SRMR    = round(SRMR, 3),
    p_value = ifelse(
      p_value < 0.001,
      "<.001",
      sprintf("%.3f", p_value)
    )
```

```r
  )

# ------------------------------------------------------------------
# Build gt table object (for HTML + image export)
# ------------------------------------------------------------------

gt_cfa_2A_summary <-
  CFA_2A_summary_print %>%
  gt::gt() %>%
  gt::cols_label(
    Model   = "Model",
    Par     = "Par",
    ChiSq   = "ChiSq",
    df      = gt::md("*df*"),
    p_value = gt::md("*p-value*"),
    CFI     = "CFI",
    SRMR    = "SRMR"
  ) %>%
  gt::tab_header(
    title = "Fit Statistics for the Two-Factor Model A"
  )
```

Note on correspondence to manuscript tables.
The fit indices reported in this table reproduce the numerical values used in the manuscript CFA results. The table formatting here is optimized for reproducibility and inspection of the analytic workflow, rather than for publication layout. The manuscript versions of these tables present the same values in a condensed, publication-formatted structure.

Render the two-factor PHQ-8 CFA Model A fit indices for all analytic groups in HTML, and export a high-resolution PNG version for inclusion in the PDF.

```r
# Render the Model A fit table in HTML
gt_cfa_2A_summary

gt::gtsave(
  data     = gt_cfa_2A_summary,
  filename = "table4_twofactorA_cfa_fit.png",
  path     = "tables",
  zoom     = 2
)
```

Insert the saved two-factor PHQ-8 CFA Model A fit table image into the PDF version of the supplement.

```r
knitr::include_graphics("tables/table4_twofactorA_cfa_fit.png")
```

Fit Statistics for the Two-Factor Model A						
Model	Par	ChiSq	*df*	*p-value*	CFI	SRMR
Total Sample	33	116.56	19	<.001	0.997	0.035
Total Men	33	78.81	19	<.001	0.994	0.049
Total Women	33	76.22	19	<.001	0.997	0.035
Filipino Americans	33	69.99	19	<.001	0.996	0.041
Filipino American Men	33	59.76	19	<.001	0.991	0.063
Filipino American Women	33	44.41	19	<.001	0.996	0.042
European Americans	33	61.94	19	<.001	0.997	0.035
European American Men	33	39.15	19	0.004	0.996	0.048
European American Women	33	42.69	19	0.001	0.998	0.036

Factor correlations for Two-factor Model A by subgroup.

Standardized correlations between the cognitive-affective and somatic factors are reported to evaluate the degree of association between the two latent dimensions across analytic subgroups. Correlations are obtained from the **lavaan** standardized two-factor solution and reflect the latent factor associations estimated using ordinal indicators. The magnitude of these correlations provides evidence regarding the distinctiveness of the cognitive-affective and somatic dimensions.

```
# -------------------------------------------------------------------
# Extract standardized latent factor correlations (AFF-SOM) for Model A
# -------------------------------------------------------------------

get_factor_corr_2A <- function(fit, sample_name) {
  cor_lv <- lavInspect(fit, "cor.lv")

  data.frame(
    Sample      = sample_name,
    Factor_1    = "AFF",
    Factor_2    = "SOM",
    Correlation = cor_lv["AFF", "SOM"],
    stringsAsFactors = FALSE
  )
}
```

```r
CFA_2A_corr <- rbind(
  get_factor_corr_2A(fit_2A_total_men,   "Total Men"),
  get_factor_corr_2A(fit_2A_total_women, "Total Women"),
  get_factor_corr_2A(fit_2A_fm,          "Filipino American Men"),
  get_factor_corr_2A(fit_2A_fw,          "Filipino American Women"),
  get_factor_corr_2A(fit_2A_em,          "European American Men"),
  get_factor_corr_2A(fit_2A_ew,          "European American Women"),
  get_factor_corr_2A(fit_2A_fa,          "Filipino Americans"),
  get_factor_corr_2A(fit_2A_ea,          "European Americans")
)

# Round correlations once, shared by HTML and PDF
CFA_2A_corr_print <- CFA_2A_corr %>%
  dplyr::mutate(Correlation = round(Correlation, 2))

# -------------------------------------------------------------------
# Build gt table object (for HTML + interactive view)
# -------------------------------------------------------------------

gt_cfa_2A_corr <- CFA_2A_corr_print %>%
  gt::gt() %>%
  gt::cols_label(
    Sample      = "Sample",
    Factor_1    = "Factor 1",
    Factor_2    = "Factor 2",
    Correlation = "Correlation"
  ) %>%
  gt::tab_header(
    title = "Standardized Factor Correlations for Two-Factor Model A of the
     ↪  PHQ-8"
  )
```

Render the two-factor PHQ-8 CFA Model A latent factor correlations for all analytic groups in HTML, and export a high-resolution PNG version for inclusion in the PDF.

```r
# HTML: show the gt table in the supplement
gt_cfa_2A_corr

# Save as supplementary CFA output (not part of a manuscript table)
gt::gtsave(
  data     = gt_cfa_2A_corr,
  filename = "cfa_twofactorA_factor_correlations.png",
  path     = "tables",
  zoom     = 2
)
```

Insert the saved two-factor PHQ-8 CFA Model A latent factor correlation table image into the PDF version of the supplement.

```
knitr::include_graphics("tables/cfa_twofactorA_factor_correlations.png")
```

Standardized Factor Correlations for Two-Factor Model A of the PHQ-8

Sample	Factor 1	Factor 2	Correlation
Total Men	AFF	SOM	0.86
Total Women	AFF	SOM	0.85
Filipino American Men	AFF	SOM	0.90
Filipino American Women	AFF	SOM	0.81
European American Men	AFF	SOM	0.85
European American Women	AFF	SOM	0.87
Filipino Americans	AFF	SOM	0.84
European Americans	AFF	SOM	0.87

Standardized factor loadings are reported to summarize the strength of association between each PHQ-8 item and its respective latent factor under the Two-factor Model A specification. Patterns of loadings across analytic subgroups are examined to assess the stability of item–factor relationships for the cognitive-affective and somatic symptom domains.

```
# -------------------------------------------------------------------
# Extract standardized factor loadings (AFF, SOM) for Model A
# -------------------------------------------------------------------

get_loadings_2A <- function(fit, group_label) {
  pe <- parameterEstimates(fit, standardized = TRUE)

  pe <- pe[
    pe$op == "=~" & pe$lhs %in% c("AFF", "SOM"),
    c("lhs", "rhs", "std.all")
  ]

  data.frame(
    Factor  = pe$lhs,
    Item    = toupper(pe$rhs),
    Group   = group_label,
```

```r
    Loading = pe$std.all,
    stringsAsFactors = FALSE
  )
}

# ----------------------------------------------------------------------
# Helper to compute omega for a single factor from a lavaan fit
# (based on standardized loadings and residual variances)
# ----------------------------------------------------------------------

compute_omega_factor_2A <- function(fit, factor_name) {
  pe <- parameterEstimates(fit, standardized = TRUE)

  # Standardized loadings for this factor
  load_rows <- pe$op == "=~" & pe$lhs == factor_name
  lambdas   <- pe$std.all[load_rows]
  items     <- pe$rhs[load_rows]

  # Residual variances for those items (std.all, lhs == rhs)
  theta_rows <- pe$op == "~~" & pe$lhs %in% items & pe$lhs == pe$rhs
  thetas     <- pe$std.all[theta_rows]

  # McDonald's omega for an equal-weight composite of the items
  num <- (sum(lambdas))^2
  den <- num + sum(thetas)

  omega <- num / den
  return(omega)
}

# ----------------------------------------------------------------------
# Collect loadings across all analytic subgroups (long format)
# ----------------------------------------------------------------------

loadings_2A_long <- rbind(
  get_loadings_2A(fit_2A_total_men,   "Total Men"),
  get_loadings_2A(fit_2A_total_women, "Total Women"),
  get_loadings_2A(fit_2A_fm,          "Filipino American Men"),
  get_loadings_2A(fit_2A_fw,          "Filipino American Women"),
  get_loadings_2A(fit_2A_em,          "European American Men"),
  get_loadings_2A(fit_2A_ew,          "European American Women"),
  get_loadings_2A(fit_2A_fa,          "Filipino Americans"),
  get_loadings_2A(fit_2A_ea,          "European Americans")
)

# ----------------------------------------------------------------------
# Compute omegas for AFF and SOM in each subgroup
# ----------------------------------------------------------------------
```

```
get_omegas_2A <- function(fit, group_label) {
  data.frame(
    Factor = c("AFF", "SOM"),
    Group  = group_label,
    Omega  = c(
      compute_omega_factor_2A(fit, "AFF"),
      compute_omega_factor_2A(fit, "SOM")
    ),
    stringsAsFactors = FALSE
  )
}

omegas_2A_long <- rbind(
  get_omegas_2A(fit_2A_total_men,   "Total Men"),
  get_omegas_2A(fit_2A_total_women, "Total Women"),
  get_omegas_2A(fit_2A_fm,          "Filipino American Men"),
  get_omegas_2A(fit_2A_fw,          "Filipino American Women"),
  get_omegas_2A(fit_2A_em,          "European American Men"),
  get_omegas_2A(fit_2A_ew,          "European American Women"),
  get_omegas_2A(fit_2A_fa,          "Filipino Americans"),
  get_omegas_2A(fit_2A_ea,          "European Americans")
)

# ---------------------------------------------------------------------
# Convert loadings to wide format and order by factor and item
# ---------------------------------------------------------------------

loadings_2A_wide <- loadings_2A_long %>%
  dplyr::mutate(Loading = round(Loading, 2)) %>%
  tidyr::pivot_wider(
    id_cols     = c(Factor, Item),
    names_from = Group,
    values_from = Loading
  ) %>%
  dplyr::arrange(Factor, Item)

# ---------------------------------------------------------------------
# Convert omegas to wide format and append as "Omega" row per factor
# ---------------------------------------------------------------------

group_cols <- setdiff(names(loadings_2A_wide), c("Factor", "Item"))

omegas_2A_wide <- omegas_2A_long %>%
  dplyr::mutate(Omega = round(Omega, 2)) %>%
  tidyr::pivot_wider(
    id_cols     = Factor,
    names_from = Group,
```

```
    values_from = Omega
  ) %>%
  dplyr::mutate(Item = "Omega") %>%
  dplyr::select(Factor, Item, dplyr::all_of(group_cols))

# Bind loadings and omega rows, keep "Omega" at bottom within each factor
loadings_2A_with_omega <- dplyr::bind_rows(loadings_2A_wide, omegas_2A_wide) %>%
  dplyr::mutate(
    Item = factor(
      Item,
      levels = c(
        "PHQ1", "PHQ2", "PHQ3", "PHQ4", "PHQ5", "PHQ6", "PHQ7", "PHQ8", "Omega"
      )
    )
  ) %>%
  dplyr::arrange(Factor, Item)

# --------------------------------------------------------------------
# Build gt table object (for HTML + interactive view)
# --------------------------------------------------------------------

gt_cfa_2A_loadings <- loadings_2A_with_omega %>%
  gt::gt() %>%
  gt::tab_spanner(
    label   = "Standardized Loadings (rows PHQ1-PHQ8) and Omega (row Omega)",
    columns = -c(Factor, Item)
  ) %>%
  gt::tab_header(
    title = gt::md("Standardized Loadings and Omega for Two-Factor Model A of the
     ↪  PHQ-8 by Group")
  )
```

Interpretation and manuscript correspondence.
The standardized factor loadings and McDonald's ω values reproduced here correspond to the estimates reported in Tables 5 and 6 of the manuscript. The tables in this supplement are presented in a code-forward format to support transparent replication of the underlying estimates. The manuscript tables present the same numerical results in a typeset publication format.

Render the two-factor PHQ-8 CFA Model A standardized factor loadings and McDonald's omega estimates for all analytic groups in HTML, and export a high-resolution PNG version for inclusion in the PDF.

```
# HTML: show loadings/omega table in the supplement
gt_cfa_2A_loadings

# Save for manuscript Tables 5 and 6
gt::gtsave(
  data     = gt_cfa_2A_loadings,
  filename = "table5_6_twofactorA_loadings_omega.png",
```

```r
  path      = "tables",
  zoom      = 2
)
```

Insert the saved two-factor PHQ-8 CFA Model A standardized loadings and McDonald's omega table image into the PDF version of the supplement.

```r
knitr::include_graphics("tables/table5_6_twofactorA_loadings_omega.png")
```

Standardized Loadings and Omega for Two-Factor Model A of the PHQ-8 by Group

		Standardized Loadings (rows PHQ1–PHQ8) and Omega (row Omega)							
Factor	Item	Total Men	Total Women	Filipino American Men	Filipino American Women	European American Men	European American Women	Filipino Americans	European Americans
AFF	PHQ1	0.82	0.85	0.81	0.84	0.83	0.86	0.83	0.86
AFF	PHQ2	0.78	0.82	0.72	0.80	0.82	0.84	0.76	0.83
AFF	PHQ6	0.86	0.82	0.91	0.81	0.83	0.83	0.85	0.83
AFF	PHQ7	0.74	0.81	0.75	0.76	0.74	0.84	0.75	0.80
AFF	PHQ8	0.78	0.71	0.77	0.67	0.78	0.74	0.70	0.75
AFF	Omega	0.90	0.90	0.89	0.89	0.90	0.91	0.89	0.91
SOM	PHQ3	0.75	0.80	0.75	0.83	0.76	0.77	0.81	0.77
SOM	PHQ4	0.79	0.81	0.75	0.86	0.82	0.77	0.84	0.80
SOM	PHQ5	0.75	0.86	0.80	0.82	0.72	0.88	0.82	0.82
SOM	Omega	0.81	0.86	0.81	0.88	0.81	0.85	0.86	0.84

5.3 Two-factor model B

The second two-factor specification represents an alternative cognitive-affective and somatic partition of PHQ-8 items proposed in prior psychometric work. This model allows direct comparison of competing two-factor structures under identical estimation and subgrouping conditions. Model fit indices and parameter estimates are presented to facilitate comparison with both the one-factor model and Two-factor Model A.

In this specification, the **cognitive-affective** factor is indicated by items 1, 2, and 6–8, whereas the **somatic** factor is indicated by items 3–5.

```r
# ------------------------------------------------------------------
# Diagram code for Two-Factor Model B
# ------------------------------------------------------------------

diagram_code_twofactor_B <- "
digraph {
```

```
  label = 'Two Factor Model B of the PHQ-8 (alternative Cognitive-Affective vs.
↪  Somatic allocation)'
  graph [ranksep = .3]

  # Observed indicators
  node [shape = box]
  phq1;
  phq2;
  phq6;

  node [shape = box]
  phq3;
  phq4;
  phq5;
  phq7;
  phq8;

  # Latent factors
  node [shape = circle]
  Affective [label = 'Cognitive-Affective'];
  Somatic;

  # Factor loadings
  edge [minlen = 3.5]

  Affective -> phq1
  Affective -> phq2
  Affective -> phq6

  Somatic -> phq3
  Somatic -> phq4
  Somatic -> phq5
  Somatic -> phq7
  Somatic -> phq8

  # Correlated factors
  Affective -> Somatic [dir = both]
}
"

# Build DiagrammeR graph object
twofactor_B_graph <- DiagrammeR::grViz(diagram_code_twofactor_B)

# ---------------------------------------------------------------------
# Ensure static PNG exists for PDF output
# ---------------------------------------------------------------------
```

```r
svg_txt <- DiagrammeRsvg::export_svg(twofactor_B_graph)
rsvg::rsvg_png(
  charToRaw(svg_txt),
  file = "figures/cfa_twofactor_modelB.png"
)

# -------------------------------------------------------------------
# Format-specific rendering
# -------------------------------------------------------------------

if (knitr::is_html_output()) {

  # HTML / IDE preview: interactive DiagrammeR widget
  twofactor_B_graph

} else if (knitr::is_latex_output()) {

  # PDF: use static PNG
  knitr::include_graphics("figures/cfa_twofactor_modelB.png")

} else {

  # Fallback for other formats / interactive cases
  twofactor_B_graph
}
```

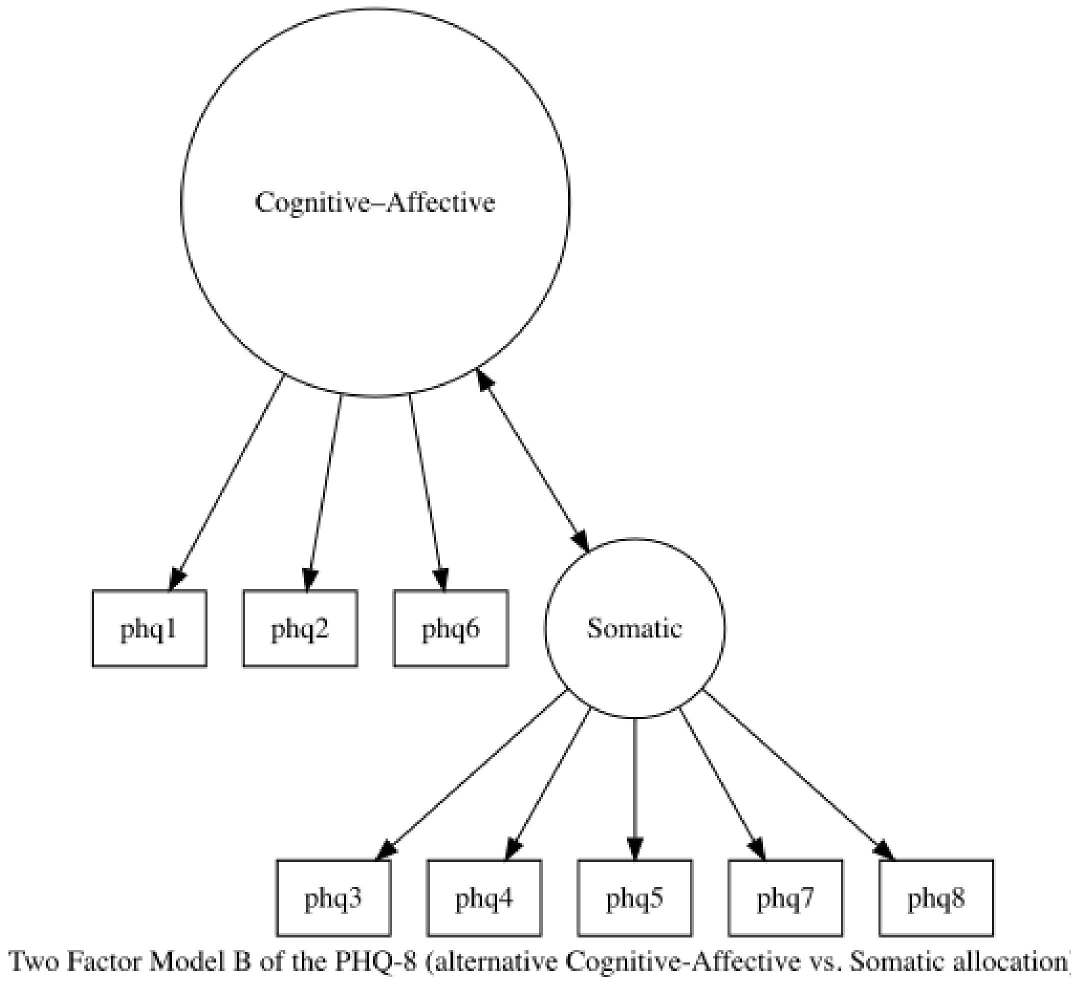


Figure 1: Diagram of the two-factor PHQ-8 Model B (alternative Cognitive-Affective vs. Somatic allocation).

Two-factor Model B CFA estimated separately across analytic subgroups.

```
# -------------------------------------------------------------------
# Two-Factor PHQ-8 CFA - Model B (WLSMV, THETA)
#   AFF = Cognitive-Affective items (phq1, phq2, phq6)
#   SOM = Somatic items  (phq3, phq4, phq5, phq7, phq8)
# -------------------------------------------------------------------

phq_cfa_2B_model <- "
  AFF =~ phq1 + phq2 + phq6
  SOM =~ phq3 + phq4 + phq5 + phq7 + phq8
"

run_cfa_2B <- function(data) {
  cfa(
    model           = phq_cfa_2B_model,
```

```r
    data             = data,
    ordered          = c('phq1','phq2','phq3','phq4',
                         'phq5','phq6','phq7','phq8'),
    estimator        = "WLSMV",
    parameterization = "theta"
  )
}

# Total analytic sample (Model B)
fit_2B_total <- run_cfa_2B(
  phq_subs %>%
    select(phq1:phq8)
)

# Subgroup models
fit_2B_total_men   <- run_cfa_2B(subset_total_men)
fit_2B_total_women <- run_cfa_2B(subset_total_women)

fit_2B_fm <- run_cfa_2B(subset_fm)
fit_2B_fw <- run_cfa_2B(subset_fw)

fit_2B_em <- run_cfa_2B(subset_em)
fit_2B_ew <- run_cfa_2B(subset_ew)

fit_2B_fa <- run_cfa_2B(subset_fa)
fit_2B_ea <- run_cfa_2B(subset_ea)
```

Model fit indices for Two-factor Model B across subgroups.

```r
# -------------------------------------------------------------------
# Collect Two-Factor Model B fit indices across analytic subgroups
# -------------------------------------------------------------------

CFA_2B_summary <- data.frame(
  Model = c(
    "Total Sample",
    "Total Men",
    "Total Women",
    "Filipino Americans",
    "Filipino American Men",
    "Filipino American Women",
    "European Americans",
    "European American Men",
    "European American Women"
  ),
  Par = c(
    fitMeasures(fit_2B_total,       "npar"),
```

```r
    fitMeasures(fit_2B_total_men,   "npar"),
    fitMeasures(fit_2B_total_women, "npar"),
    fitMeasures(fit_2B_fa,          "npar"),
    fitMeasures(fit_2B_fm,          "npar"),
    fitMeasures(fit_2B_fw,          "npar"),
    fitMeasures(fit_2B_ea,          "npar"),
    fitMeasures(fit_2B_em,          "npar"),
    fitMeasures(fit_2B_ew,          "npar")
  ),
  ChiSq = c(
    fitMeasures(fit_2B_total,       "chisq"),
    fitMeasures(fit_2B_total_men,   "chisq"),
    fitMeasures(fit_2B_total_women, "chisq"),
    fitMeasures(fit_2B_fa,          "chisq"),
    fitMeasures(fit_2B_fm,          "chisq"),
    fitMeasures(fit_2B_fw,          "chisq"),
    fitMeasures(fit_2B_ea,          "chisq"),
    fitMeasures(fit_2B_em,          "chisq"),
    fitMeasures(fit_2B_ew,          "chisq")
  ),
  df = c(
    fitMeasures(fit_2B_total,       "df"),
    fitMeasures(fit_2B_total_men,   "df"),
    fitMeasures(fit_2B_total_women, "df"),
    fitMeasures(fit_2B_fa,          "df"),
    fitMeasures(fit_2B_fm,          "df"),
    fitMeasures(fit_2B_fw,          "df"),
    fitMeasures(fit_2B_ea,          "df"),
    fitMeasures(fit_2B_em,          "df"),
    fitMeasures(fit_2B_ew,          "df")
  ),
  p_value = c(
    fitMeasures(fit_2B_total,       "pvalue"),
    fitMeasures(fit_2B_total_men,   "pvalue"),
    fitMeasures(fit_2B_total_women, "pvalue"),
    fitMeasures(fit_2B_fa,          "pvalue"),
    fitMeasures(fit_2B_fm,          "pvalue"),
    fitMeasures(fit_2B_fw,          "pvalue"),
    fitMeasures(fit_2B_ea,          "pvalue"),
    fitMeasures(fit_2B_em,          "pvalue"),
    fitMeasures(fit_2B_ew,          "pvalue")
  ),
  CFI = c(
    fitMeasures(fit_2B_total,       "cfi"),
    fitMeasures(fit_2B_total_men,   "cfi"),
    fitMeasures(fit_2B_total_women, "cfi"),
    fitMeasures(fit_2B_fa,          "cfi"),
    fitMeasures(fit_2B_fm,          "cfi"),
```

```r
    fitMeasures(fit_2B_fw,          "cfi"),
    fitMeasures(fit_2B_ea,          "cfi"),
    fitMeasures(fit_2B_em,          "cfi"),
    fitMeasures(fit_2B_ew,          "cfi")
  ),
  SRMR = c(
    fitMeasures(fit_2B_total,       "srmr"),
    fitMeasures(fit_2B_total_men,   "srmr"),
    fitMeasures(fit_2B_total_women, "srmr"),
    fitMeasures(fit_2B_fa,          "srmr"),
    fitMeasures(fit_2B_fm,          "srmr"),
    fitMeasures(fit_2B_fw,          "srmr"),
    fitMeasures(fit_2B_ea,          "srmr"),
    fitMeasures(fit_2B_em,          "srmr"),
    fitMeasures(fit_2B_ew,          "srmr")
  ),
  stringsAsFactors = FALSE
)

# ------------------------------------------------------------------
# Apply rounding and p-value formatting once (shared by HTML / PDF)
# ------------------------------------------------------------------

CFA_2B_summary_print <-
  CFA_2B_summary %>%
  dplyr::mutate(
    ChiSq   = round(ChiSq, 2),
    CFI     = round(CFI, 3),
    SRMR    = round(SRMR, 3),
    p_value = ifelse(
      p_value < 0.001,
      "<.001",
      sprintf("%.3f", p_value)
    )
  )

# ------------------------------------------------------------------
# Build gt table object (for HTML + interactive view)
# ------------------------------------------------------------------

gt_cfa_2B_summary <-
  CFA_2B_summary_print %>%
  gt::gt() %>%
  gt::cols_label(
    Model   = "Model",
    Par     = "Par",
    ChiSq   = "ChiSq",
    df      = gt::md("*df*"),
```

```
    p_value = gt::md("*p-value*"),
    CFI     = "CFI",
    SRMR    = "SRMR"
  ) %>%
  gt::tab_header(
    title = "Fit Statistics for the Two-Factor Model B"
  )
```

> **Note on correspondence to manuscript tables.**
> The fit indices reported in this table reproduce the numerical values used in the manuscript CFA results. The table formatting here is optimized for reproducibility and inspection of the analytic workflow, rather than for publication layout. The manuscript versions of these tables present the same values in a condensed, publication-formatted structure.

Render the two-factor PHQ-8 CFA Model B fit indices for all analytic groups in HTML, and export a high-resolution PNG version for inclusion in the PDF.

```
# Render the Model B fit table in HTML
gt_cfa_2B_summary

# Also save the PNG for manuscript Table 4 (Two-Factor Model B panel)
gt::gtsave(
  data     = gt_cfa_2B_summary,
  filename = "table4_twofactorB_cfa_fit.png",
  path     = "tables",
  zoom     = 2
)
```

Insert the saved two-factor PHQ-8 CFA Model B fit table image into the PDF version of the supplement.

```
knitr::include_graphics("tables/table4_twofactorB_cfa_fit.png")
```

Model	Par	ChiSq	*df*	*p-value*	CFI	SRMR
Total Sample	33	112.81	19	<.001	0.997	0.035
Total Men	33	33.21	19	0.023	0.998	0.032
Total Women	33	98.14	19	<.001	0.996	0.041
Filipino Americans	33	74.86	19	<.001	0.995	0.044
Filipino American Men	33	23.80	19	0.204	0.999	0.041
Filipino American Women	33	70.06	19	<.001	0.993	0.055
European Americans	33	48.27	19	<.001	0.998	0.031
European American Men	33	21.32	19	0.319	1.000	0.036
European American Women	33	40.36	19	0.003	0.998	0.035

Fit Statistics for the Two-Factor Model B

Factor correlations for Two-factor Model B by subgroup.

Standardized correlations between the cognitive-affective and somatic factors are reported to evaluate the degree of association between the two latent dimensions under the Two-factor Model B specification across analytic subgroups. Correlations are obtained from the **lavaan** standardized two-factor solution and reflect the latent factor associations estimated using ordinal indicators. The magnitude of these correlations informs the extent to which the cognitive-affective and somatic dimensions are empirically distinct.

```
# -------------------------------------------------------------------
# Extract standardized latent factor correlations (AFF-SOM) for Model B
# -------------------------------------------------------------------

get_factor_corr_2B <- function(fit, sample_name) {
  cor_lv <- lavInspect(fit, "cor.lv")

  data.frame(
    Sample      = sample_name,
    Factor_1    = "AFF",
    Factor_2    = "SOM",
    Correlation = cor_lv["AFF", "SOM"],
    stringsAsFactors = FALSE
  )
}
```

```r
CFA_2B_corr <- rbind(
  get_factor_corr_2B(fit_2B_total_men,   "Total Men"),
  get_factor_corr_2B(fit_2B_total_women, "Total Women"),
  get_factor_corr_2B(fit_2B_fm,          "Filipino American Men"),
  get_factor_corr_2B(fit_2B_fw,          "Filipino American Women"),
  get_factor_corr_2B(fit_2B_em,          "European American Men"),
  get_factor_corr_2B(fit_2B_ew,          "European American Women"),
  get_factor_corr_2B(fit_2B_fa,          "Filipino Americans"),
  get_factor_corr_2B(fit_2B_ea,          "European Americans")
)

# Round correlations once, shared across formats
CFA_2B_corr_print <- CFA_2B_corr %>%
  dplyr::mutate(Correlation = round(Correlation, 2))

# -------------------------------------------------------------------
# Build gt table object (for HTML + interactive view)
# -------------------------------------------------------------------

gt_cfa_2B_corr <- CFA_2B_corr_print %>%
  gt::gt() %>%
  gt::cols_label(
    Sample      = "Sample",
    Factor_1    = "Factor 1",
    Factor_2    = "Factor 2",
    Correlation = "Correlation"
  ) %>%
  gt::tab_header(
    title = "Standardized Factor Correlations for Two-Factor Model B of the
     ↪  PHQ-8"
  )
```

Render the two-factor PHQ-8 CFA Model B latent factor correlations for all analytic groups in HTML, and export a high-resolution PNG version for inclusion in the PDF.

```r
# HTML: show factor correlation table inline
gt_cfa_2B_corr

# Save supplementary CFA PNG
gt::gtsave(
  data     = gt_cfa_2B_corr,
  filename = "cfa_twofactorB_factor_correlations.png",
  path     = "tables",
  zoom     = 2
)
```

Insert the saved two-factor PHQ-8 CFA Model B latent factor correlation table image into the PDF version of the supplement.

```
knitr::include_graphics("tables/cfa_twofactorB_factor_correlations.png")
```

Standardized Factor Correlations for Two-Factor Model B of the PHQ-8			
Sample	Factor 1	Factor 2	Correlation
Total Men	AFF	SOM	0.83
Total Women	AFF	SOM	0.87
Filipino American Men	AFF	SOM	0.82
Filipino American Women	AFF	SOM	0.85
European American Men	AFF	SOM	0.83
European American Women	AFF	SOM	0.88
Filipino Americans	AFF	SOM	0.85
European Americans	AFF	SOM	0.87

Standardized factor loadings are reported to summarize the strength of association between each PHQ-8 item and its assigned latent factor under the Two-factor Model B specification. Comparisons of loading patterns across analytic subgroups and across the competing two-factor models provide information about the stability and interpretability of alternative cognitive-affective and somatic partitions.

```
# ------------------------------------------------------------------
# Extract standardized factor loadings (AFF, SOM) for Model B
# ------------------------------------------------------------------

get_loadings_2B <- function(fit, group_label) {
  pe <- parameterEstimates(fit, standardized = TRUE)

  pe <- pe[
    pe$op == "=~" & pe$lhs %in% c("AFF", "SOM"),
    c("lhs", "rhs", "std.all")
  ]

  data.frame(
    Factor  = pe$lhs,
    Item    = toupper(pe$rhs),
    Group   = group_label,
    Loading = pe$std.all,
    stringsAsFactors = FALSE
```

```r
  )
}

# ------------------------------------------------------------------
# Helper to compute omega for a single factor from a lavaan fit
# (based on standardized loadings and residual variances)
# ------------------------------------------------------------------

compute_omega_factor_2B <- function(fit, factor_name) {
  pe <- parameterEstimates(fit, standardized = TRUE)

  # Standardized loadings for this factor
  load_rows <- pe$op == "=~" & pe$lhs == factor_name
  lambdas   <- pe$std.all[load_rows]
  items     <- pe$rhs[load_rows]

  # Residual variances for those items (std.all, lhs == rhs)
  theta_rows <- pe$op == "~~" & pe$lhs %in% items & pe$lhs == pe$rhs
  thetas     <- pe$std.all[theta_rows]

  # McDonald's omega for an equal-weight composite
  num <- (sum(lambdas))^2
  den <- num + sum(thetas)

  omega <- num / den
  return(omega)
}

# ------------------------------------------------------------------
# Collect loadings across all analytic subgroups (long format)
# ------------------------------------------------------------------

loadings_2B_long <- rbind(
  get_loadings_2B(fit_2B_total_men,   "Total Men"),
  get_loadings_2B(fit_2B_total_women, "Total Women"),
  get_loadings_2B(fit_2B_fm,          "Filipino American Men"),
  get_loadings_2B(fit_2B_fw,          "Filipino American Women"),
  get_loadings_2B(fit_2B_em,          "European American Men"),
  get_loadings_2B(fit_2B_ew,          "European American Women"),
  get_loadings_2B(fit_2B_fa,          "Filipino Americans"),
  get_loadings_2B(fit_2B_ea,          "European Americans")
)

# ------------------------------------------------------------------
# Compute omegas for AFF and SOM in each subgroup
# ------------------------------------------------------------------

get_omegas_2B <- function(fit, group_label) {
```

```r
  data.frame(
    Factor = c("AFF", "SOM"),
    Group  = group_label,
    Omega  = c(
      compute_omega_factor_2B(fit, "AFF"),
      compute_omega_factor_2B(fit, "SOM")
    ),
    stringsAsFactors = FALSE
  )
}

omegas_2B_long <- rbind(
  get_omegas_2B(fit_2B_total_men,   "Total Men"),
  get_omegas_2B(fit_2B_total_women, "Total Women"),
  get_omegas_2B(fit_2B_fm,          "Filipino American Men"),
  get_omegas_2B(fit_2B_fw,          "Filipino American Women"),
  get_omegas_2B(fit_2B_em,          "European American Men"),
  get_omegas_2B(fit_2B_ew,          "European American Women"),
  get_omegas_2B(fit_2B_fa,          "Filipino Americans"),
  get_omegas_2B(fit_2B_ea,          "European Americans")
)

# --------------------------------------------------------------------
# Convert loadings to wide format and order by factor and item
# --------------------------------------------------------------------

loadings_2B_wide <- loadings_2B_long %>%
  dplyr::mutate(Loading = round(Loading, 2)) %>%
  tidyr::pivot_wider(
    id_cols     = c(Factor, Item),
    names_from = Group,
    values_from = Loading
  ) %>%
  dplyr::arrange(Factor, Item)

# --------------------------------------------------------------------
# Convert omegas to wide format and append as "Omega" row per factor
# --------------------------------------------------------------------

group_cols <- setdiff(names(loadings_2B_wide), c("Factor", "Item"))

omegas_2B_wide <- omegas_2B_long %>%
  dplyr::mutate(Omega = round(Omega, 2)) %>%
  tidyr::pivot_wider(
    id_cols     = Factor,
    names_from = Group,
    values_from = Omega
  ) %>%
```

```
  # Add Item label and align column order with loadings table
  dplyr::mutate(Item = "Omega") %>%
  dplyr::select(Factor, Item, dplyr::all_of(group_cols))

# Bind loadings and omega rows, keep "Omega" at bottom within each factor
loadings_2B_with_omega <- dplyr::bind_rows(loadings_2B_wide, omegas_2B_wide) %>%
  dplyr::mutate(
    Item = factor(
      Item,
      levels = c(
        "PHQ1", "PHQ2", "PHQ3", "PHQ4", "PHQ5", "PHQ6", "PHQ7", "PHQ8", "Omega"
      )
    )
  ) %>%
  dplyr::arrange(Factor, Item)

# ------------------------------------------------------------------
# Format loadings + omega table for reporting (Tables 5 and 6)
# ------------------------------------------------------------------

gt_cfa_2B_loadings <- loadings_2B_with_omega %>%
  gt::gt() %>%
  gt::tab_spanner(
    label   = "Standardized Loadings (rows PHQ1-PHQ8) and Omega (row Omega)",
    columns = -c(Factor, Item)
  ) %>%
  gt::tab_header(
    title = gt::md("Standardized Loadings and Omega for Two-Factor Model B of the
    ↪  PHQ-8 by Group")
  )
```

Interpretation and manuscript correspondence.
The standardized factor loadings and McDonald's ω values reproduced here correspond to the estimates reported in Tables 5 and 6 of the manuscript. The tables in this supplement are presented in a code-forward format to support transparent replication of the underlying estimates. The manuscript tables present the same numerical results in a typeset publication format.

Render the two-factor PHQ-8 CFA Model B standardized factor loadings and McDonald's omega estimates for all analytic groups in HTML, and export a high-resolution PNG version for inclusion in the PDF.

```
# HTML: show loadings/omega table inline
gt_cfa_2B_loadings

# Save for manuscript Tables 5 and 6 (PNG in tables/ directory)
gt::gtsave(
  data     = gt_cfa_2B_loadings,
  filename = "table5_6_twofactorB_loadings_omega.png",
  path     = "tables",
```

```r
  zoom      = 2
)
```

Insert the saved two-factor PHQ-8 CFA Model B standardized loadings and McDonald's omega table image into the PDF version of the supplement.

```r
knitr::include_graphics("tables/table5_6_twofactorB_loadings_omega.png")
```

Standardized Loadings and Omega for Two-Factor Model B of the PHQ-8 by Group

		Standardized Loadings (rows PHQ1–PHQ8) and Omega (row Omega)							
Factor	Item	Total Men	Total Women	Filipino American Men	Filipino American Women	European American Men	European American Women	Filipino Americans	European Americans
AFF	PHQ1	0.84	0.86	0.82	0.86	0.85	0.88	0.84	0.87
AFF	PHQ2	0.81	0.84	0.75	0.81	0.85	0.86	0.78	0.85
AFF	PHQ6	0.89	0.84	0.95	0.83	0.85	0.85	0.87	0.85
AFF	Omega	0.88	0.89	0.88	0.87	0.89	0.90	0.87	0.89
SOM	PHQ3	0.73	0.77	0.74	0.80	0.73	0.75	0.79	0.75
SOM	PHQ4	0.76	0.78	0.74	0.83	0.78	0.75	0.81	0.77
SOM	PHQ5	0.73	0.82	0.79	0.79	0.69	0.84	0.80	0.79
SOM	PHQ7	0.76	0.82	0.77	0.76	0.76	0.86	0.75	0.82
SOM	PHQ8	0.80	0.71	0.80	0.67	0.80	0.75	0.70	0.76
SOM	Omega	0.87	0.89	0.88	0.88	0.87	0.89	0.88	0.88

---

6. Measurement invariance analyses

> **Correspondence to manuscript:** Fit indices and ΔCFI/ΔSRMR results in this section reproduce the multigroup measurement invariance analyses reported in Table 7 of the manuscript.

Measurement invariance was evaluated using multi-group confirmatory factor analysis to assess whether PHQ-8 measurement properties were comparable across sex within selected ethnic groups. Invariance testing proceeded sequentially from configural to metric and scalar models, allowing increasingly restrictive equality constraints to be evaluated under identical estimation conditions.

All invariance models were estimated in **lavaan** using an estimator appropriate for ordinal indicators (DWLS / WLSMV-style). Model fit indices and changes in fit were examined to evaluate the extent to which equality constraints were supported. These analyses were conducted to assess the comparability of latent factor interpretation across groups rather than to test substantive mean differences.

Invariance decisions were evaluated using ΔCFI and ΔSRMR thresholds recommended for categorical indicators under WLSMV estimation (Chen, 2007).

Configural, metric, and scalar invariance models estimated across sex.

Invariance models are estimated sequentially, beginning with a configural model and proceeding to metric and scalar constraints. This sequence allows evaluation of whether factor loadings and item thresholds can be considered equivalent across sex within each population.

ℹ Syntax scaffolding: PHQ-8 measurement invariance in MplusAutomation

- **Goal.** These chunks estimate **multi-group CFA models in Mplus** to evaluate configural, metric, and scalar invariance across ethnicity and sex, reproducing the fit indices and LR tests described in the manuscript.
- **Data and grouping.**
  - The `VARIABLE` block in each `mplusObject()` specifies:
    - ∗ `NAMES = ...` and `USEVARIABLES` for PHQ-8 items and grouping variables (e.g., `filipino`, `female`).
    - ∗ `CATEGORICAL ARE phq1-phq8;` so that items are treated as ordinal.
    - ∗ `GROUPING IS filipino (0 = European 1 = Filipino);` or `GROUPING IS female (0 = male 1 = female);` to define the comparison groups.
- **Estimation settings.**
  - The `ANALYSIS` block sets

```
ESTIMATOR = WLSMV;
PARAMETERIZATION = THETA;
```

    mirroring the lavaan settings for ordinal CFA and ensuring consistency with the main article and LTA workflow.
- **Model structure across invariance steps.**
  - **Configural models** (`ethnicity_configural`, `sex_configural`) specify the PHQ-8 factor structure separately by group without equality constraints.
  - **Metric models** add equality constraints on factor loadings across groups.
  - **Scalar models** additionally constrain thresholds (and, where appropriate, intercepts) across groups.
    These constraints are implemented through the `MODEL` and `MODEL CONSTRAINT` sections of the Mplus syntax embedded in the `mplusObject()`.
- **Running Mplus via R.**
  - `MplusAutomation::mplusModeler()` writes the `.inp` file and calls Mplus to run each model (e.g., `Ethnicity_Configural.inp`, `Ethnicity_Metric.inp`, `Ethnicity_Scalar.inp`).
  - `MplusAutomation::readModels()` reads the resulting `.out` files back into R, giving access to χ^2, df, CFI, TLI, RMSEA, and SRMR.
- **Summarizing fit and invariance comparisons.**
  - Fit indices for each model (configural, metric, scalar) are extracted from the `parameters` and `summaries` elements of the `readModels()` output.

- These are combined into small invariance summary tables using base `data.frame()` and `dplyr` operations, and formatted for the supplement using `gt::gt()`, mirroring the manuscript's invariance reporting.

- **Software dependencies.**
  - **MplusAutomation**: `mplusObject()`, `mplusModeler()`, `readModels()`.
  - **gt** and **dplyr** for compact invariance summary tables.

6.1 Invariance testing by ethnicity

Ethnicity prep: ordered items, group factor

```
# 1. Define PHQ-8 item names
phq_items <- paste0("phq", 1:8)

# 2. Ensure PHQ-8 items are treated as ordered factors for WLSMV
phq_total[phq_items] <- lapply(phq_total[phq_items], ordered)

# 3. Create ethnicity grouping factor for lavaan (European vs Filipino)
phq_total$eth_group <- factor(
  phq_total$filipino,
  levels = c(0, 1),
  labels = c("European", "Filipino")
)

# 4. One-factor PHQ-8 model
phq_model <- "
  dep =~ phq1 + phq2 + phq3 + phq4 +
         phq5 + phq6 + phq7 + phq8
"
```

Ethnicity Configural Invariance

```
# -------------------------------------------------------------------
# Configural invariance model (European vs Filipino)
#   - freely estimated loadings & thresholds across groups
# -------------------------------------------------------------------

fit_eth_config <- cfa(
  model             = phq_model,
  data              = phq_total,
  group             = "eth_group",
  ordered           = phq_items,
  estimator         = "WLSMV",
  parameterization = "theta"
```

```
)

# Extract selected fit indices (includes SRMR)
fitMeasures(
  fit_eth_config,
  c(
    "chisq", "df", "pvalue",
    "cfi", "tli",
    "rmsea", "rmsea.ci.lower", "rmsea.ci.upper",
    "srmr"
  )
)
```

Ethnicity Metric Invariance

```
# ------------------------------------------------------------------
# Metric invariance model (European vs Filipino)
#   - factor loadings constrained equal across groups
# ------------------------------------------------------------------

fit_eth_metric <- cfa(
  model            = phq_model,
  data             = phq_total,
  group            = "eth_group",
  ordered          = phq_items,
  estimator        = "WLSMV",
  parameterization = "theta",
  group.equal      = c("loadings")
)

# Extract selected fit indices (includes SRMR)
fitMeasures(
  fit_eth_metric,
  c(
    "chisq", "df", "pvalue",
    "cfi", "tli",
    "rmsea", "rmsea.ci.lower", "rmsea.ci.upper",
    "srmr"
  )
)
```

Ethnicity Scalar Invariance

```
# ------------------------------------------------------------------
# Scalar invariance model (European vs Filipino)
#   - factor loadings and thresholds constrained equal across groups
```

```r
# -----------------------------------------------------------------

fit_eth_scalar <- cfa(
  model            = phq_model,
  data             = phq_total,
  group            = "eth_group",
  ordered          = phq_items,
  estimator        = "WLSMV",
  parameterization = "theta",
  group.equal      = c("loadings", "thresholds")
)

# Extract selected fit indices (includes SRMR)
fitMeasures(
  fit_eth_scalar,
  c(
    "chisq", "df", "pvalue",
    "cfi", "tli",
    "rmsea", "rmsea.ci.lower", "rmsea.ci.upper",
    "srmr"
  )
)
```

6.2 Invariance testing by sex

Sex prep: group factor

This section defines the sex grouping variable used for multi-group invariance testing and applies the same PHQ-8 item set and model specification used in the subgroup CFAs. The resulting grouped dataset is used to estimate the configural, metric, and scalar invariance models across men and women within each population subgroup.

```r
# This chunk assumes:
# - phq_items has already been defined as phq1:phq8
# - phq_total[phq_items] has already been coerced to ordered
# - phq_model has already been defined as the one-factor PHQ-8 model

# Create sex grouping factor for lavaan (Men vs Women)
phq_total$sex_group <- factor(
  phq_total$female,
  levels = c(0, 1),
  labels = c("Men", "Women")
)
```

Sex Configural Invariance

```r
# -------------------------------------------------------------------
# Configural invariance model (Male vs Female)
#   - freely estimated loadings & thresholds across sex groups
# -------------------------------------------------------------------

fit_sex_config <- cfa(
  model            = phq_model,
  data             = phq_total,
  group            = "sex_group",
  ordered          = phq_items,
  estimator        = "WLSMV",
  parameterization = "theta"
)

# Extract selected fit indices (includes SRMR)
fitMeasures(
  fit_sex_config,
  c(
    "chisq", "df", "pvalue",
    "cfi", "tli",
    "rmsea", "rmsea.ci.lower", "rmsea.ci.upper",
    "srmr"
  )
)
```

Sex Metric Invariance

```r
# -------------------------------------------------------------------
# Metric invariance model (Male vs Female)
#   - factor loadings constrained equal across sex groups
# -------------------------------------------------------------------

fit_sex_metric <- cfa(
  model            = phq_model,
  data             = phq_total,
  group            = "sex_group",
  ordered          = phq_items,
  estimator        = "WLSMV",
  parameterization = "theta",
  group.equal      = c("loadings")
)

# Extract selected fit indices (includes SRMR)
fitMeasures(
  fit_sex_metric,
  c(
    "chisq", "df", "pvalue",
```

```
    "cfi", "tli",
    "rmsea", "rmsea.ci.lower", "rmsea.ci.upper",
    "srmr"
  )
)
```

Sex Scalar Invariance

```
# ------------------------------------------------------------------
# Scalar invariance model (Male vs Female)
#   - factor loadings and thresholds constrained equal across groups
# ------------------------------------------------------------------

fit_sex_scalar <- cfa(
  model            = phq_model,
  data             = phq_total,
  group            = "sex_group",
  ordered          = phq_items,
  estimator        = "WLSMV",
  parameterization = "theta",
  group.equal      = c("loadings", "thresholds")
)

# Extract selected fit indices (includes SRMR)
fitMeasures(
  fit_sex_scalar,
  c(
    "chisq", "df", "pvalue",
    "cfi", "tli",
    "rmsea", "rmsea.ci.lower", "rmsea.ci.upper",
    "srmr"
  )
)
```

6.3 Invariance testing among Filipino American women and men

Prep: subset to Filipino only + sex grouping

```
# 1. Subset to Filipino Americans only (filipino == 1)
phq_fil <- subset(phq_total, filipino == 1)

# 2. Ensure PHQ-8 items are ordered factors within the subset
phq_fil[phq_items] <- lapply(phq_fil[phq_items], ordered)

# 3. Create sex grouping factor within Filipinos
#    (0 = Male, 1 = Female)
```

```r
phq_fil$sex_group <- factor(
  phq_fil$female,
  levels = c(0, 1),
  labels = c("Male", "Female")
)
```

Filipino Sex Configural Invariance

```r
# ------------------------------------------------------------------
# Configural invariance (Filipino subsample: Male vs Female)
#   - freely estimated loadings & thresholds across sex groups
# ------------------------------------------------------------------

fit_fil_sex_config <- cfa(
  model            = phq_model,
  data             = phq_fil,
  group            = "sex_group",
  ordered          = phq_items,
  estimator        = "WLSMV",
  parameterization = "theta"
)

# Extract selected fit indices (includes SRMR)
fitMeasures(
  fit_fil_sex_config,
  c(
    "chisq", "df", "pvalue",
    "cfi", "tli",
    "rmsea", "rmsea.ci.lower", "rmsea.ci.upper",
    "srmr"
  )
)
```

Filipino Sex Metric Invariance

```r
# ------------------------------------------------------------------
# Metric invariance (Filipino subsample: Male vs Female)
#   - factor loadings constrained equal across sex groups
# ------------------------------------------------------------------

fit_fil_sex_metric <- cfa(
  model            = phq_model,
  data             = phq_fil,
  group            = "sex_group",
  ordered          = phq_items,
  estimator        = "WLSMV",
```

```r
  parameterization = "theta",
  group.equal      = c("loadings")
)

# Extract selected fit indices (includes SRMR)
fitMeasures(
  fit_fil_sex_metric,
  c(
    "chisq", "df", "pvalue",
    "cfi", "tli",
    "rmsea", "rmsea.ci.lower", "rmsea.ci.upper",
    "srmr"
  )
)
```

Filipino Sex Scalar Invariance

```r
# --------------------------------------------------------------------
# Scalar invariance (Filipino subsample: Male vs Female)
#   - factor loadings and thresholds constrained equal across sex
# --------------------------------------------------------------------

fit_fil_sex_scalar <- cfa(
  model            = phq_model,
  data             = phq_fil,
  group            = "sex_group",
  ordered          = phq_items,
  estimator        = "WLSMV",
  parameterization = "theta",
  group.equal      = c("loadings", "thresholds")
)

# Extract selected fit indices (includes SRMR)
fitMeasures(
  fit_fil_sex_scalar,
  c(
    "chisq", "df", "pvalue",
    "cfi", "tli",
    "rmsea", "rmsea.ci.lower", "rmsea.ci.upper",
    "srmr"
  )
)
```

6.4 Invariance testing among European American women and men

Prep: subset to American / European only + sex grouping

```r
# 1. Subset to European Americans only (filipino == 0)
phq_eur <- subset(phq_total, filipino == 0)

# 2. Ensure PHQ-8 items are ordered factors within the subset
phq_eur[phq_items] <- lapply(phq_eur[phq_items], ordered)

# 3. Create sex grouping factor within Europeans
#    (0 = Male, 1 = Female)
phq_eur$sex_group <- factor(
  phq_eur$female,
  levels = c(0, 1),
  labels = c("Male", "Female")
)
```

European American Women and Men Configural

```r
# -------------------------------------------------------------------
# Configural invariance (European subsample: Male vs Female)
#   - freely estimated loadings & thresholds across sex groups
# -------------------------------------------------------------------

fit_eur_sex_config <- cfa(
  model            = phq_model,
  data             = phq_eur,
  group            = "sex_group",
  ordered          = phq_items,
  estimator        = "WLSMV",
  parameterization = "theta"
)

# Extract selected fit indices (includes SRMR)
fitMeasures(
  fit_eur_sex_config,
  c(
    "chisq", "df", "pvalue",
    "cfi", "tli",
    "rmsea", "rmsea.ci.lower", "rmsea.ci.upper",
    "srmr"
  )
)
```

European American Women and Men Metric

```r
# -------------------------------------------------------------------
# Metric invariance (European subsample: Male vs Female)
#   - factor loadings constrained equal across sex groups
```

```r
# -----------------------------------------------------------------

fit_eur_sex_metric <- cfa(
  model           = phq_model,
  data            = phq_eur,
  group           = "sex_group",
  ordered         = phq_items,
  estimator       = "WLSMV",
  parameterization = "theta",
  group.equal     = c("loadings")
)

# Extract selected fit indices (includes SRMR)
fitMeasures(
  fit_eur_sex_metric,
  c(
    "chisq", "df", "pvalue",
    "cfi", "tli",
    "rmsea", "rmsea.ci.lower", "rmsea.ci.upper",
    "srmr"
  )
)
```

European American Women and Men Scalar

```r
# -----------------------------------------------------------------
# Scalar invariance (European subsample: Male vs Female)
#   - factor loadings and thresholds constrained equal across sex
# -----------------------------------------------------------------

fit_eur_sex_scalar <- cfa(
  model           = phq_model,
  data            = phq_eur,
  group           = "sex_group",
  ordered         = phq_items,
  estimator       = "WLSMV",
  parameterization = "theta",
  group.equal     = c("loadings", "thresholds")
)

# Extract selected fit indices (includes SRMR)
fitMeasures(
  fit_eur_sex_scalar,
  c(
    "chisq", "df", "pvalue",
    "cfi", "tli",
    "rmsea", "rmsea.ci.lower", "rmsea.ci.upper",
```

```
    "srmr"
  )
)
```

Construction of Fit Index Extraction Utility Function

```
# Construct a single summary row for a given model:
#   - Group, Model label
#   - Chi-square, df, CFI, SRMR
#   - Optional ΔCFI, ΔSRMR relative to a reference (typically configural)

get_fit_row <- function(fit, group_label, model_label, ref_fit = NULL) {
  fm <- fitMeasures(fit, c("chisq", "df", "cfi", "srmr"))

  row <- data.frame(
    Group = group_label,
    Model = model_label,
    ChiSq = as.numeric(fm["chisq"]),
    df    = as.numeric(fm["df"]),
    CFI   = as.numeric(fm["cfi"]),
    SRMR  = as.numeric(fm["srmr"]),
    stringsAsFactors = FALSE
  )

  if (!is.null(ref_fit)) {
    ref_fm <- fitMeasures(ref_fit, c("cfi", "srmr"))
    row$Delta_CFI  <- row$CFI  - as.numeric(ref_fm["cfi"])
    row$Delta_SRMR <- row$SRMR - as.numeric(ref_fm["srmr"])
  } else {
    row$Delta_CFI  <- NA_real_
    row$Delta_SRMR <- NA_real_
  }

  row
}
```

Compilation of Model Fit Statistics Across Invariance Models and Subgroups

```
# -------------------------------------------------------------------
# Build stacked table of configural / metric / scalar models
# across all invariance analyses, with optional ΔCFI / ΔSRMR
# -------------------------------------------------------------------

invar_table <- bind_rows(
```

```
  # Ethnicity - full sample
  get_fit_row(fit_eth_config,  "Ethnicity (Filipino vs European)", "Configural"),
  get_fit_row(fit_eth_metric,  "Ethnicity (Filipino vs European)", "Metric",
              ref_fit = fit_eth_config),
  get_fit_row(fit_eth_scalar,  "Ethnicity (Filipino vs European)", "Scalar",
              ref_fit = fit_eth_metric),

  # sex - full sample
  get_fit_row(fit_sex_config, "sex (All participants)", "Configural"),
  get_fit_row(fit_sex_metric, "sex (All participants)", "Metric",
              ref_fit = fit_sex_config),
  get_fit_row(fit_sex_scalar, "sex (All participants)", "Scalar",
              ref_fit = fit_sex_metric),

  # sex - Filipino subsample
  get_fit_row(fit_fil_sex_config, "sex (Filipino Americans)", "Configural"),
  get_fit_row(fit_fil_sex_metric, "sex (Filipino Americans)", "Metric",
              ref_fit = fit_fil_sex_config),
  get_fit_row(fit_fil_sex_scalar, "sex (Filipino Americans)", "Scalar",
              ref_fit = fit_fil_sex_metric),

  # sex - European subsample
  get_fit_row(fit_eur_sex_config, "sex (European Americans)", "Configural"),
  get_fit_row(fit_eur_sex_metric, "sex (European Americans)", "Metric",
              ref_fit = fit_eur_sex_config),
  get_fit_row(fit_eur_sex_scalar, "sex (European Americans)", "Scalar",
              ref_fit = fit_eur_sex_metric)
)

# -------------------------------------------------------------------
# Round and format numeric columns for presentation
# -------------------------------------------------------------------

invar_table <- invar_table %>%
  mutate(
    ChiSq      = round(ChiSq, 2),
    CFI        = round(CFI, 3),
    SRMR       = round(SRMR, 3),
    Delta_CFI  = round(Delta_CFI, 3),
    Delta_SRMR = round(Delta_SRMR, 3)
  ) %>%
  mutate(
    Delta_CFI  = ifelse(is.na(Delta_CFI),  "-", sprintf("%.3f", Delta_CFI)),
    Delta_SRMR = ifelse(is.na(Delta_SRMR), "-", sprintf("%.3f", Delta_SRMR))
  )
```

Formatted Presentation of Measurement Invariance Model Fit Results

```r
# ---------------------------------------------------------------------
# Format invariance fit table (Ethnicity + sex invariance models)
# ---------------------------------------------------------------------

gt_invariance <- invar_table %>%
  gt::gt() %>%
  gt::cols_label(
    Group      = "Grouping variable",
    Model      = "Model",
    ChiSq      = "ChiSq",
    df         = gt::md("*df*"),
    CFI        = "CFI",
    SRMR       = "SRMR",
    Delta_CFI  = "ΔCFI",
    Delta_SRMR = "ΔSRMR"
  ) %>%
  gt::cols_align(
    align   = "left",
    columns = c(Group, Model)
  ) %>%
  gt::cols_align(
    align   = "center",
    columns = c(ChiSq, df, CFI, SRMR, Delta_CFI, Delta_SRMR)
  ) %>%
  gt::tab_header(
    title = "Model Fit Statistics for One-Factor Measurement Invariance Models"
  )
```

Render the multigroup PHQ-8 measurement invariance model fit indices (configural, metric, and scalar) for all group comparisons in HTML, and export a high-resolution PNG version for inclusion in the PDF.

```r
# Render invariance fit table in HTML
gt_invariance

# Save PNG for manuscript Table 7
gt::gtsave(
  data     = gt_invariance,
  filename = "table7_invariance_fit.png",
  path     = "tables",
  zoom     = 2
)
```

Insert the saved multigroup PHQ-8 measurement invariance fit table image into the PDF version of the supplement.

```r
knitr::include_graphics("tables/table7_invariance_fit.png")
```

Model Fit Statistics for One-Factor Measurement Invariance Models							
Grouping variable	Model	ChiSq	*df*	CFI	SRMR	ΔCFI	ΔSRMR
Ethnicity (Filipino vs European)	Configural	278.37	40	0.992	0.052	—	—
Ethnicity (Filipino vs European)	Metric	313.70	47	0.991	0.055	-0.001	0.003
Ethnicity (Filipino vs European)	Scalar	309.49	55	0.991	0.052	0.000	-0.003
sex (All participants)	Configural	296.34	40	0.991	0.055	—	—
sex (All participants)	Metric	340.57	47	0.989	0.059	-0.001	0.004
sex (All participants)	Scalar	309.63	55	0.991	0.055	0.001	-0.004
sex (Filipino Americans)	Configural	180.53	40	0.988	0.066	—	—
sex (Filipino Americans)	Metric	221.67	47	0.985	0.071	-0.003	0.005
sex (Filipino Americans)	Scalar	197.84	55	0.988	0.066	0.003	-0.005
sex (European Americans)	Configural	149.16	40	0.993	0.053	—	—
sex (European Americans)	Metric	183.64	47	0.992	0.059	-0.002	0.006
sex (European Americans)	Scalar	160.59	55	0.994	0.053	0.002	-0.006

---

7. Theoretically Related Variables

> **Correspondence to manuscript:** Correlation estimates in this section reproduce the associations between PHQ-8 scores and theoretically related constructs reported in Table 8 of the manuscript.

Correlations were computed between PHQ-8 mean scores and a set of theoretically related constructs (depressive symptoms, anxiety, perceived stress, brooding rumination, self-deprecation, positive self-esteem, and quality of life). Correlations were estimated for the total sample, by sex, and separately within Filipino American and European American subgroups using pairwise-complete observations. These analyses were conducted to evaluate convergent and discriminant validity of the PHQ-8 across groups.

Syntax scaffolding: Correlational validity with related constructs

- **Goal.** This chunk reproduces the **correlations between PHQ-8 total scores and theoretically related constructs** (e.g., CES-D, anxiety, perceived stress, brooding rumination, self-deprecation) for the total sample and subgroup breakdowns reported in Table 8.
- **Scales and variables.**
  - `phq_total` (or equivalent) is the summed or averaged PHQ-8 score created earlier in the workflow.
  - Additional variables (e.g., `cesd_total`, `anx_total`, `stress_total`, `brooding_total`, `selfdep_total`) are derived from the corresponding item sets using earlier scoring code.
- **Correlation computation (base R).**
  - For each analytic group (e.g., total sample, Filipino Americans, European Americans, men, women), the data are subset using either:
    - Logical filters (e.g., `subset(phq_subs, Fil_Amer == 1)`) or
    - Pre-computed subgroup data frames.
  - Correlations are computed with **base R's** `cor()` function:

    cor_matrix <- cor(subset_data[, c("phq_total", related_vars)], use = "pairwise.complete.obs")
  - where `related_vars` is a vector of the theoretically related scale scores.
- **Organization and labeling.**
  - Row and column names are set to human-readable labels for PHQ-8 and each related construct.
  - Correlation matrices are rounded (e.g., `round(cor_matrix, 2)`) for reporting.
- **Tabling.**
  - Each subgroup correlation matrix is converted to a `data.frame` and passed to **`gt::gt()`**.
  - gt functions such as `gt::tab_header()` and `gt::fmt_number()` are used to:
    - Add subgroup-specific titles (e.g., "Correlations between PHQ-8 and related constructs: Filipino Americans"),
    - Control decimal places, and
    - Ensure consistent formatting across HTML and PDF.
- **Software dependencies.**
  - **Base R**: `cor()`, `subset()`, indexing and `round()`.
  - **gt**: to create the final correlation tables for the supplement.

```
# -------------------------------------------------------------------
# PHQ-8 vs. theoretically related constructs (Table 8)
# -------------------------------------------------------------------

related_vars <- c(
```

```
  "cesd_mean",
  "gad_mean",
  "stress_mean",
  "brood_mean",
  "selfdep_mean",
  "posesteem_mean",
  "qol"
)

related_labels <- c(
  "CES-D-10",
  "Anxiety",
  "Perceived stress",
  "Brooding rumination",
  "Self-deprecation",
  "Positive esteem",
  "Quality of life"
)

# Helper: correlation of phq_mean with a given variable, by subset
compute_r <- function(data, y_var) {
  cor(
    data[["phq_mean"]],
    data[[y_var]],
    use = "pairwise.complete.obs"
  )
}

# Construct correlation table across total / sex / ethnicity cells
corr_df <- data.frame(
  Related_Variable = related_labels,
  Total_Both  = sapply(related_vars, function(v) compute_r(phq_total, v)),
  Total_Men   = sapply(related_vars, function(v) compute_r(subset(phq_total,
   ↪ female == 0), v)),
  Total_Women = sapply(related_vars, function(v) compute_r(subset(phq_total,
   ↪ female == 1), v)),
  Fil_Total   = sapply(related_vars, function(v) compute_r(subset(phq_total,
   ↪ filipino == 1), v)),
  Fil_Men     = sapply(related_vars, function(v) compute_r(subset(phq_total,
   ↪ filipino == 1 & female == 0), v)),
  Fil_Women   = sapply(related_vars, function(v) compute_r(subset(phq_total,
   ↪ filipino == 1 & female == 1), v)),
  Eur_Total   = sapply(related_vars, function(v) compute_r(subset(phq_total,
   ↪ filipino == 0), v)),
  Eur_Men     = sapply(related_vars, function(v) compute_r(subset(phq_total,
   ↪ filipino == 0 & female == 0), v)),
  Eur_Women   = sapply(related_vars, function(v) compute_r(subset(phq_total,
   ↪ filipino == 0 & female == 1), v)),
```

```
  check.names = FALSE
)

# Round correlations for display
corr_df[, -1] <- round(corr_df[, -1], 2)

# ---------------------------------------------------------------------
# Format correlation table with gt
# ---------------------------------------------------------------------

gt_phq_corr <- corr_df %>%
  gt::gt() %>%
  gt::cols_label(
    Related_Variable = "Related Variable",
    Total_Both       = "Total Sample\nBoth",
    Total_Men        = "Total Sample\nMen",
    Total_Women      = "Total Sample\nWomen",
    Fil_Total        = "Filipino Americans\nTotal",
    Fil_Men          = "Filipino Americans\nMen",
    Fil_Women        = "Filipino Americans\nWomen",
    Eur_Total        = "European Americans\nTotal",
    Eur_Men          = "European Americans\nMen",
    Eur_Women        = "European Americans\nWomen"
  ) %>%
  gt::cols_align(
    align   = "left",
    columns = Related_Variable
  ) %>%
  gt::cols_align(
    align   = "center",
    columns = -Related_Variable
  ) %>%
  gt::tab_header(
    title = "Correlations of PHQ-8 with Theoretically Related Constructs"
  )
```

Render the correlations between PHQ-8 scores and theoretically related constructs for all analytic groups in HTML, and export a high-resolution PNG version for inclusion in the PDF.

```
# Render correlation table in HTML
gt_phq_corr

# Save PNG for manuscript Table 8
gt::gtsave(
  data     = gt_phq_corr,
  filename = "table8_phq_related_correlations.png",
  path     = "tables",
```

```
  zoom      = 2
)
```

Insert the saved PHQ-8 correlations with theoretically related constructs table image into the PDF version of the supplement.

```
knitr::include_graphics("tables/table8_phq_related_correlations.png")
```

Correlations of PHQ-8 with Theoretically Related Constructs

Related Variable	Total Sample Both	Total Sample Men	Total Sample Women	Filipino Americans Total	Filipino Americans Men	Filipino Americans Women	European Americans Total	European Americans Men	European Americans Women
CES-D-10	0.76	0.74	0.77	0.72	0.73	0.72	0.79	0.74	0.81
Anxiety	0.74	0.70	0.75	0.72	0.73	0.69	0.76	0.68	0.79
Perceived stress	0.47	0.32	0.54	0.44	0.31	0.49	0.50	0.33	0.56
Brooding rumination	0.54	0.42	0.59	0.50	0.39	0.54	0.56	0.44	0.61
Self-deprecation	0.63	0.63	0.63	0.57	0.58	0.55	0.67	0.66	0.67
Positive esteem	-0.49	-0.50	-0.48	-0.41	-0.43	-0.41	-0.54	-0.55	-0.52
Quality of life	-0.52	-0.50	-0.53	-0.50	-0.47	-0.51	-0.52	-0.51	-0.53

Reproducibility metadata.
Analyses were reproduced using R (version 4.5.1; R Core Team, 2025), lavaan (version 0.6-20; Rosseel, 2012), RStudio Pro (2025), and Quarto (version 1.7; 2024). Rendering this Quarto document regenerates all models, tables, and figures shown in the supplement.

References for Software and Analytic FrameworkReferences for Software and Analytic Framework

Rosseel, Y. (2012). lavaan: An R package for structural equation modeling. *Journal of Statistical Software, 48*(2), 1–36.
R Core Team. (2025). *R: A language and environment for statistical computing.*
Posit Software, PBC. (2025). *RStudio Pro.*
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